

BFS Traversal

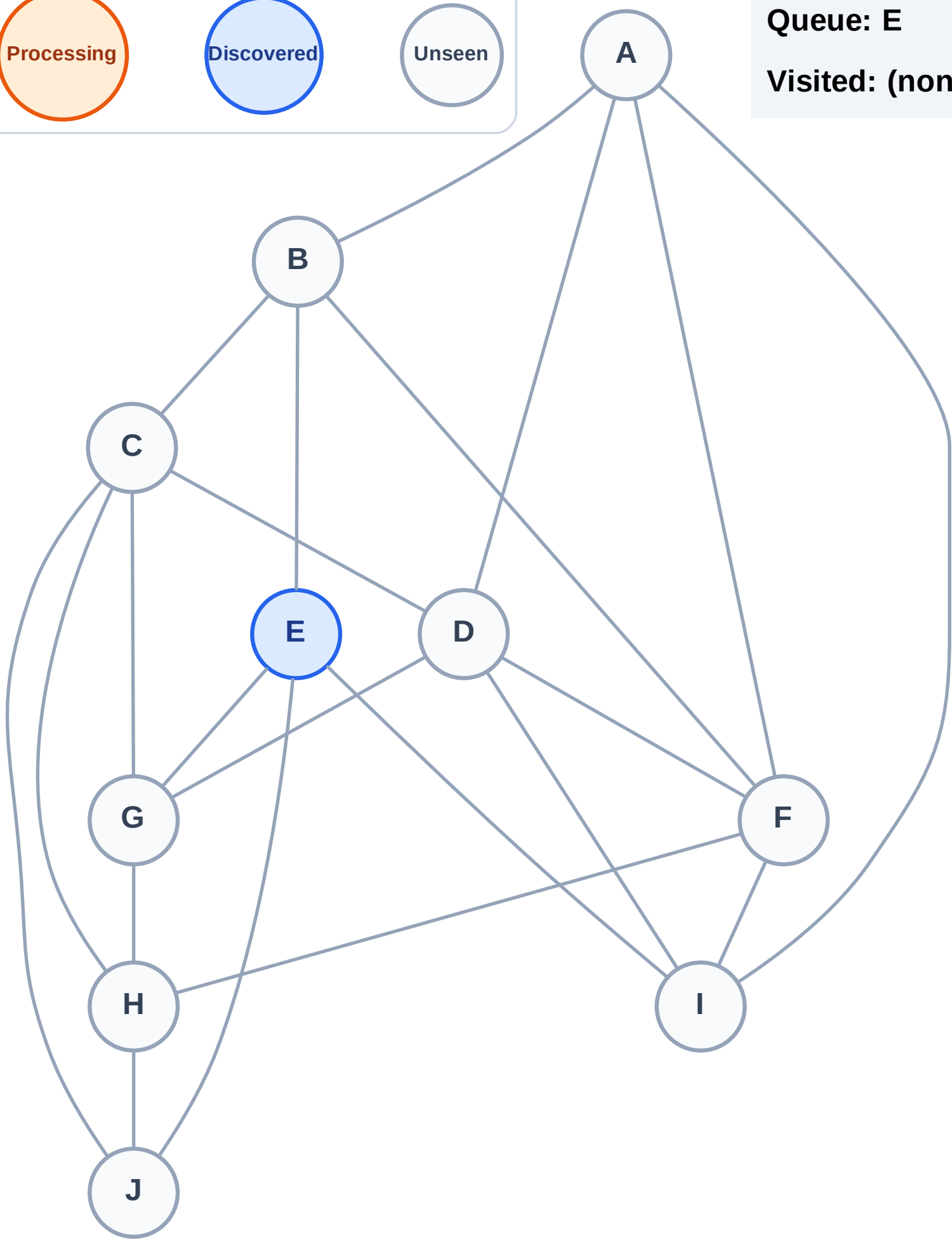
Step 1: Start at E

Legend



Queue: E

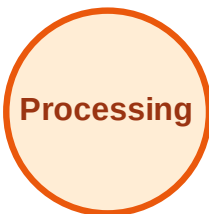
Visited: (none)



BFS Traversal

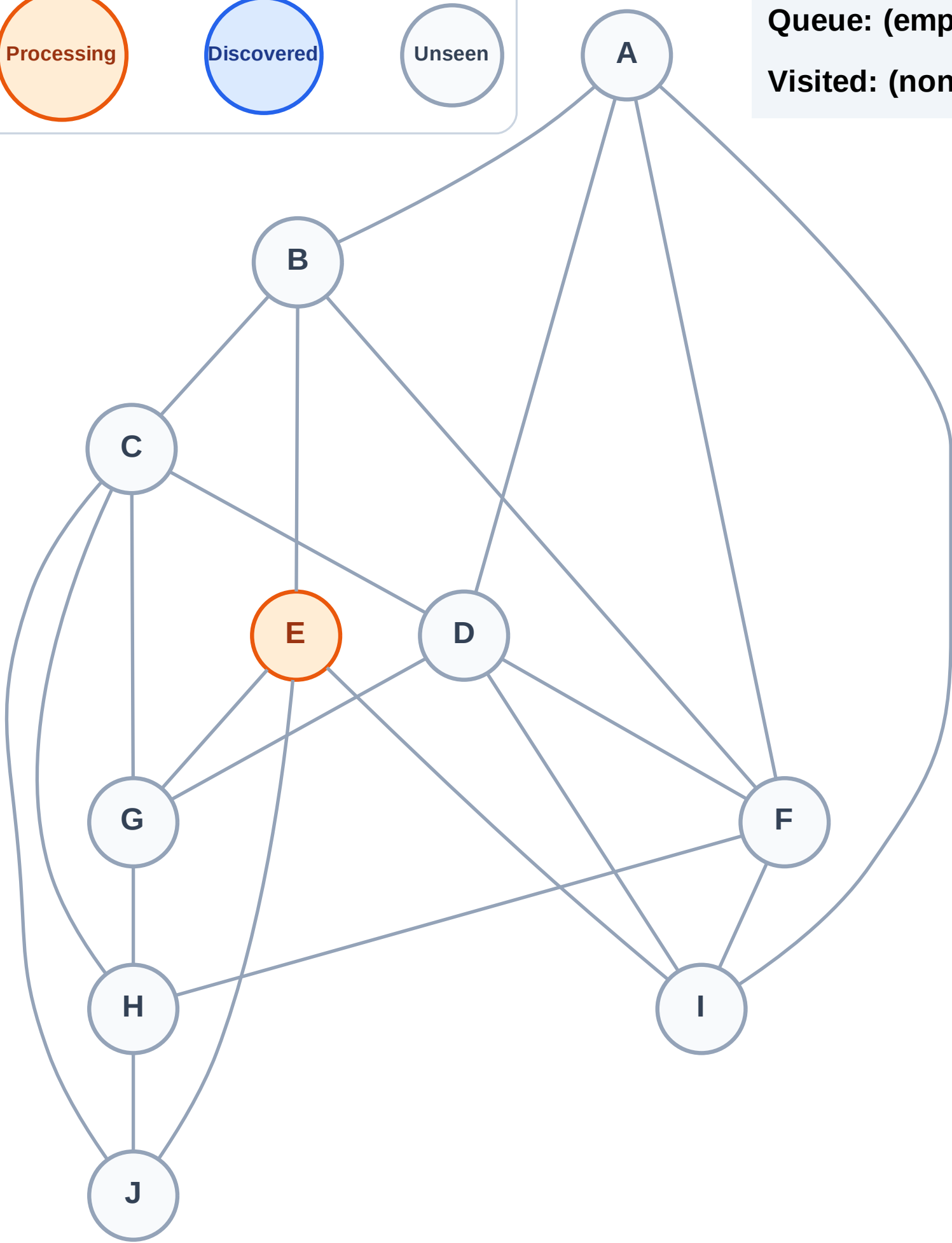
Step 2: Process node E

Legend



Queue: (empty)

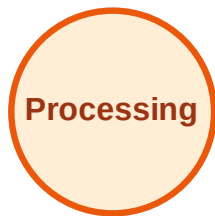
Visited: (none)



BFS Traversal

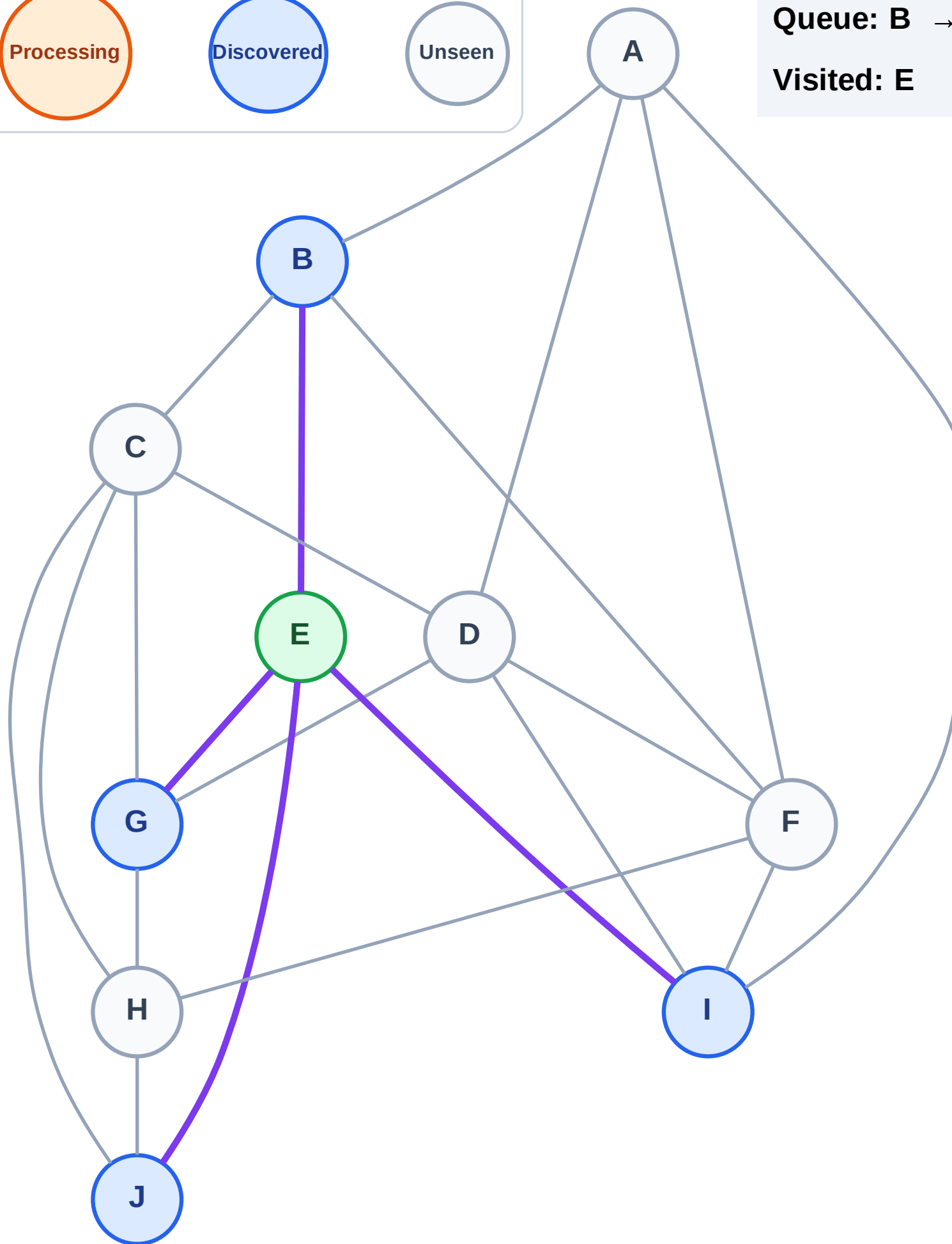
Step 3: Discover B, G, I, J from E

Legend



Queue: B → G → I → J

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

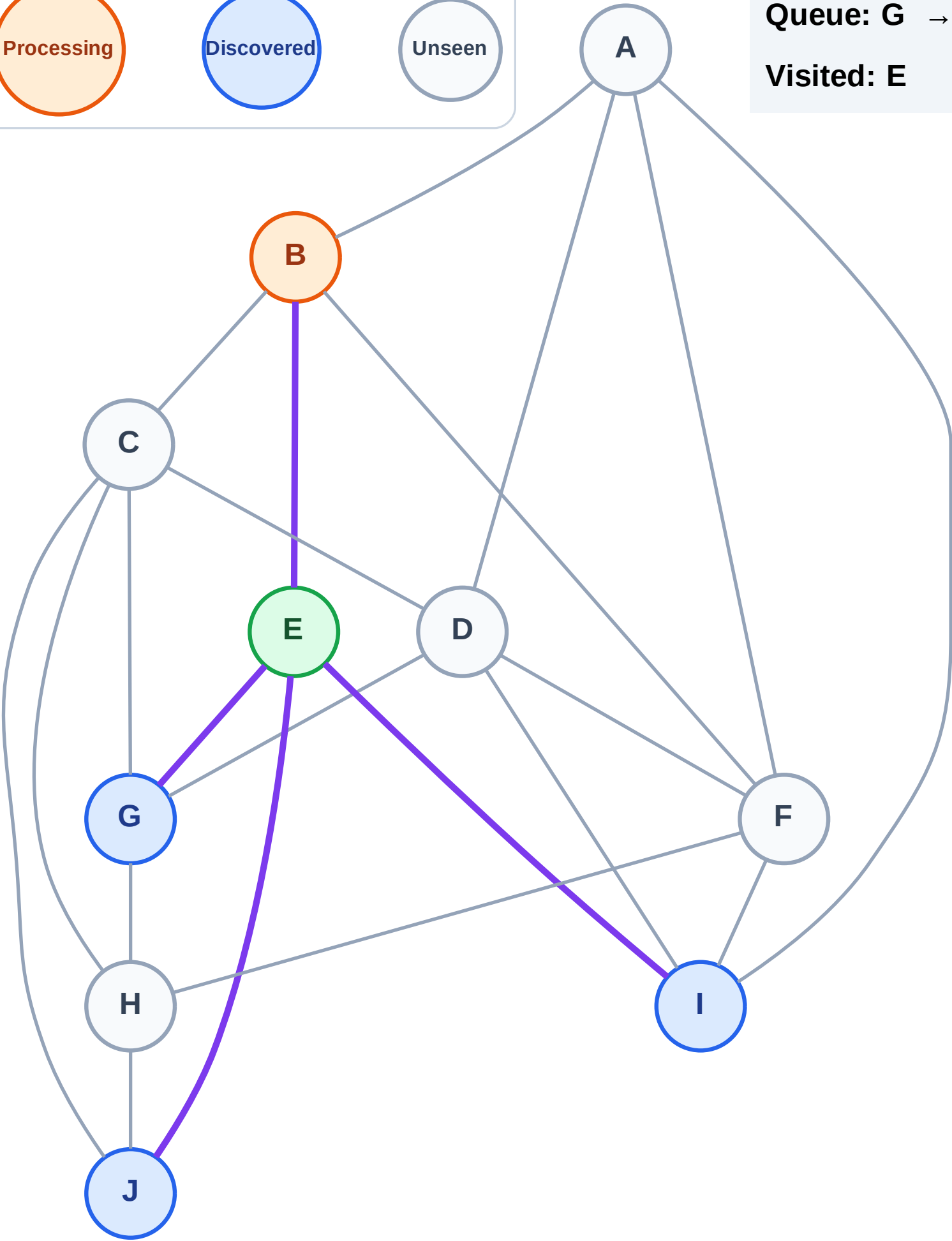
Processing

Discovered

Unseen

Queue: G → I → J

Visited: E



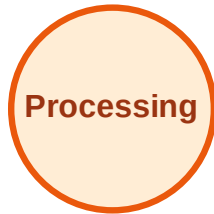
BFS Traversal

Step 5: Discover A, C, F from B

Legend



Complete



Processing

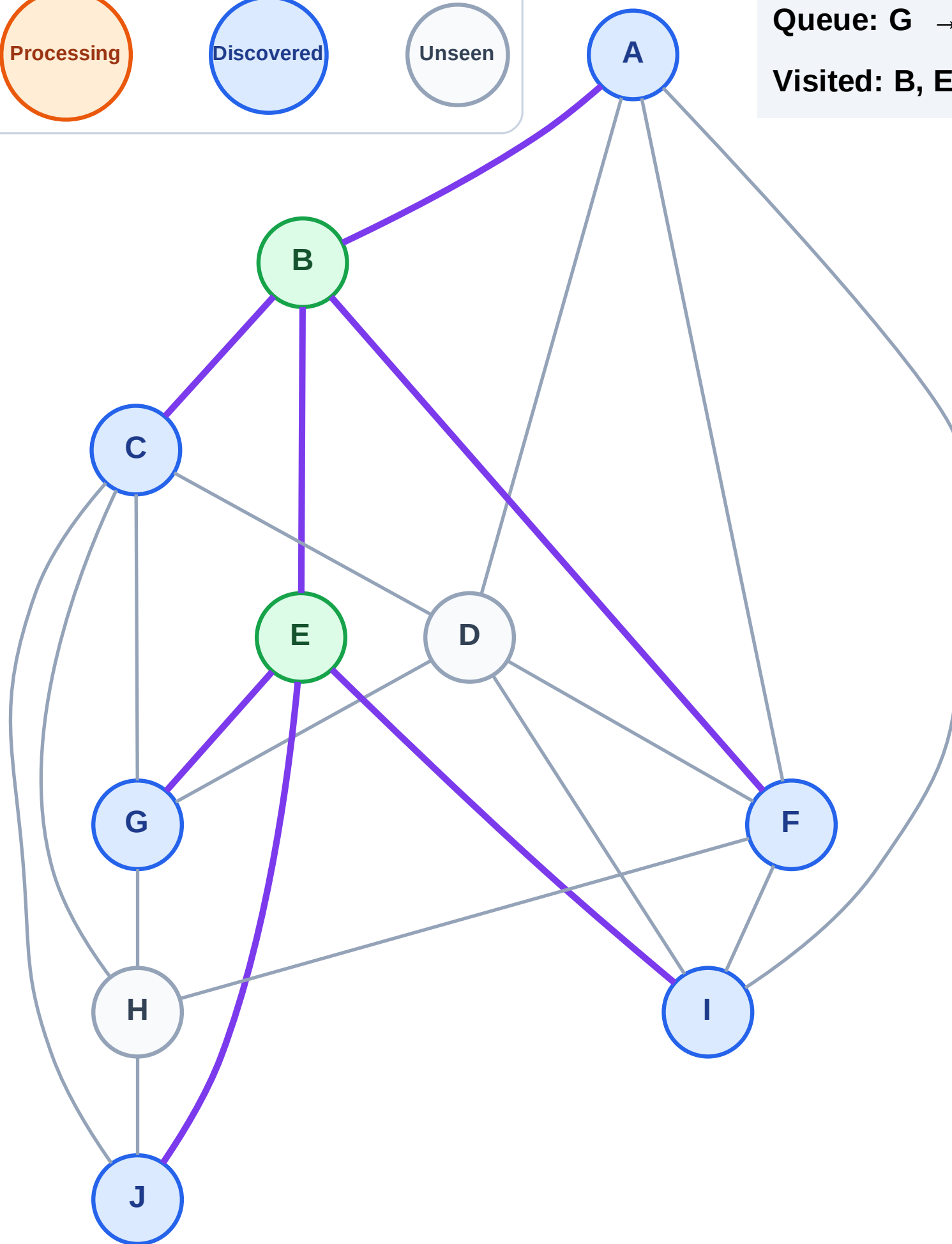


Discovered



Unseen

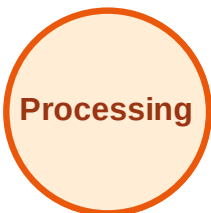
Queue: G → I → J → A → C → F
Visited: B, E



BFS Traversal

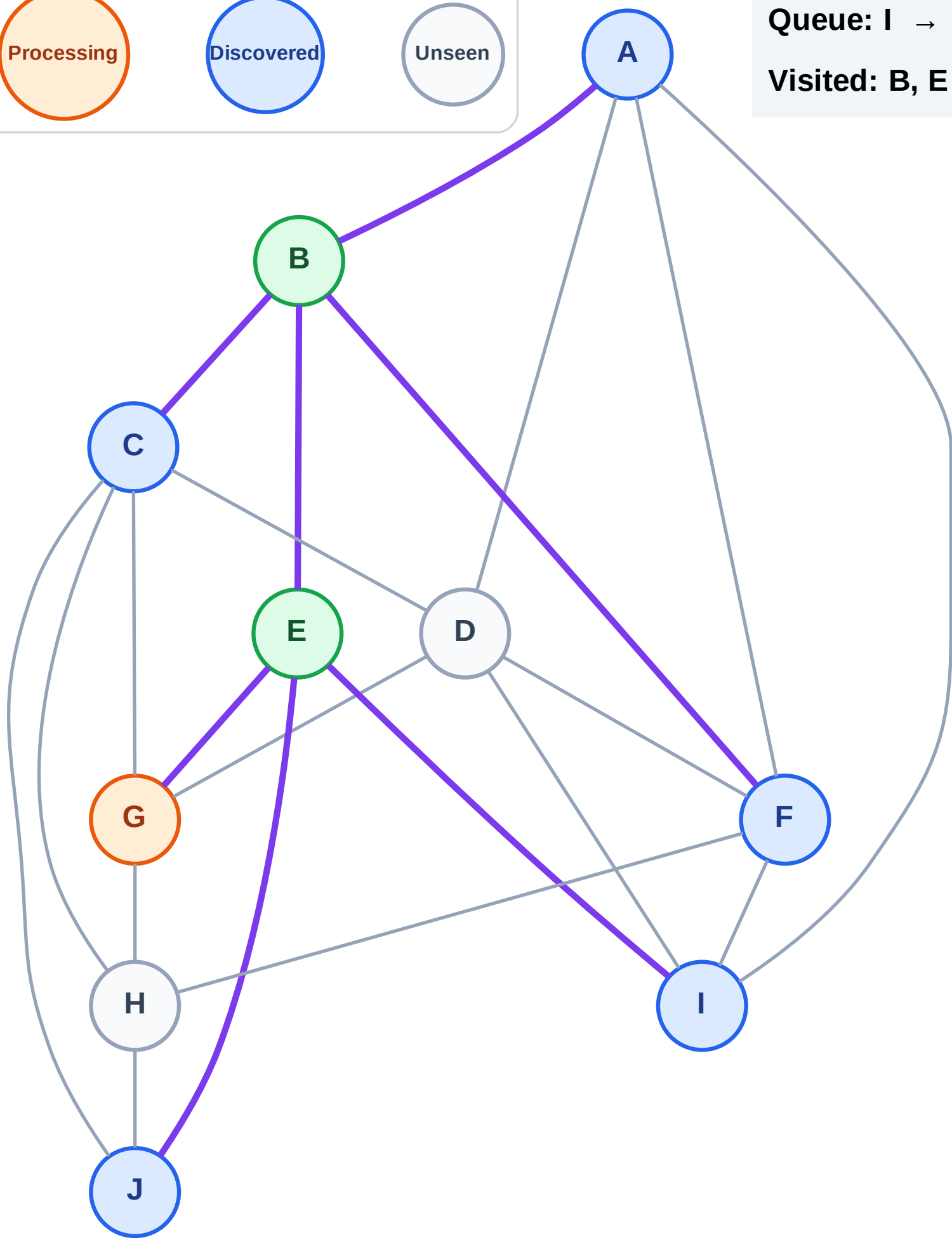
Step 6: Process node G

Legend



Queue: I → J → A → C → F

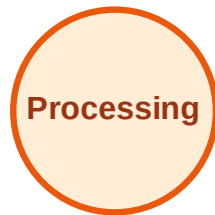
Visited: B, E



BFS Traversal

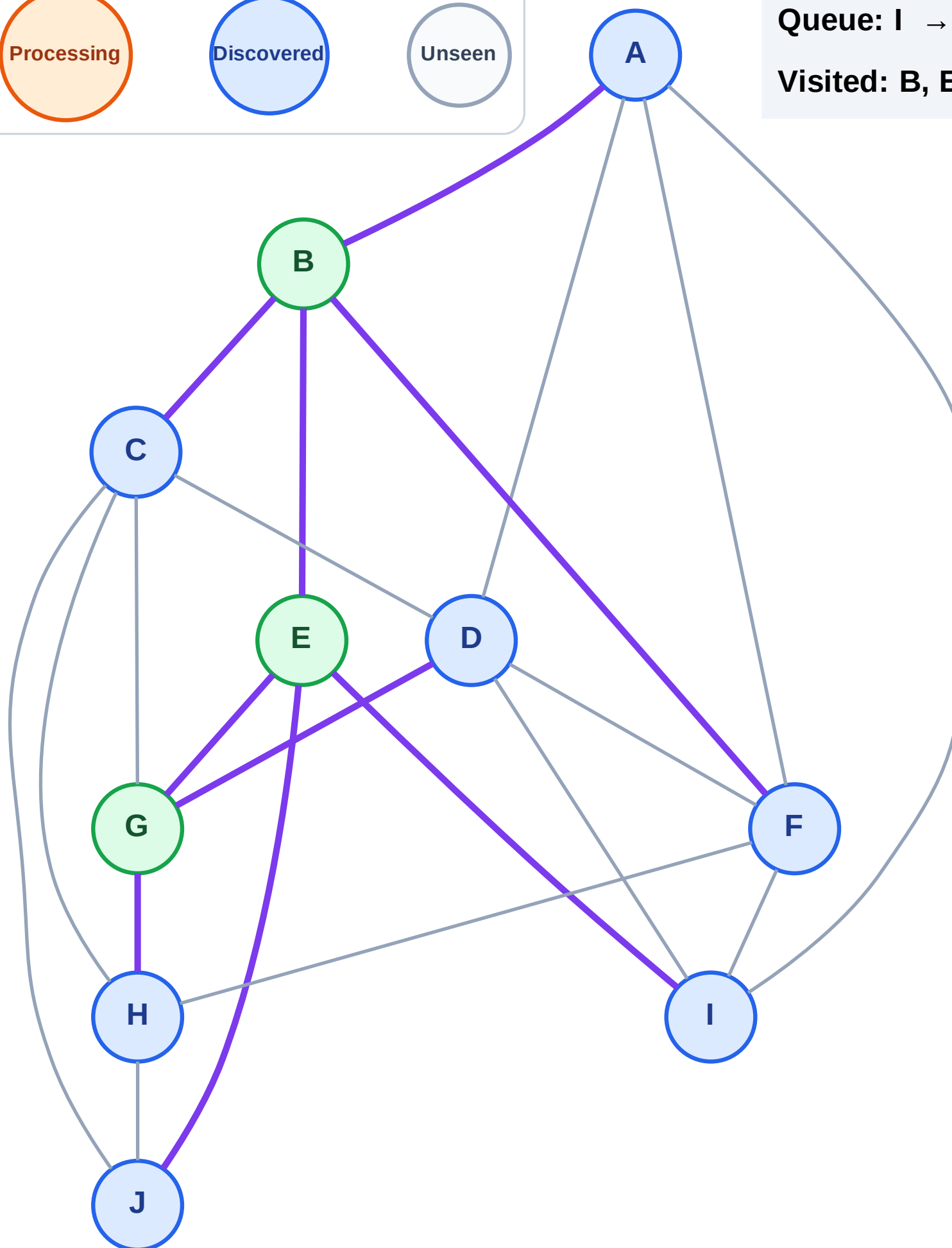
Step 7: Discover D, H from G

Legend



Queue: I → J → A → C → F → D → H

Visited: B, E, G



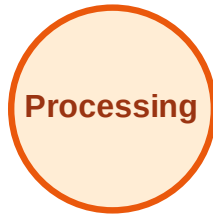
BFS Traversal

Step 8: Process node I

Legend



Complete



Processing



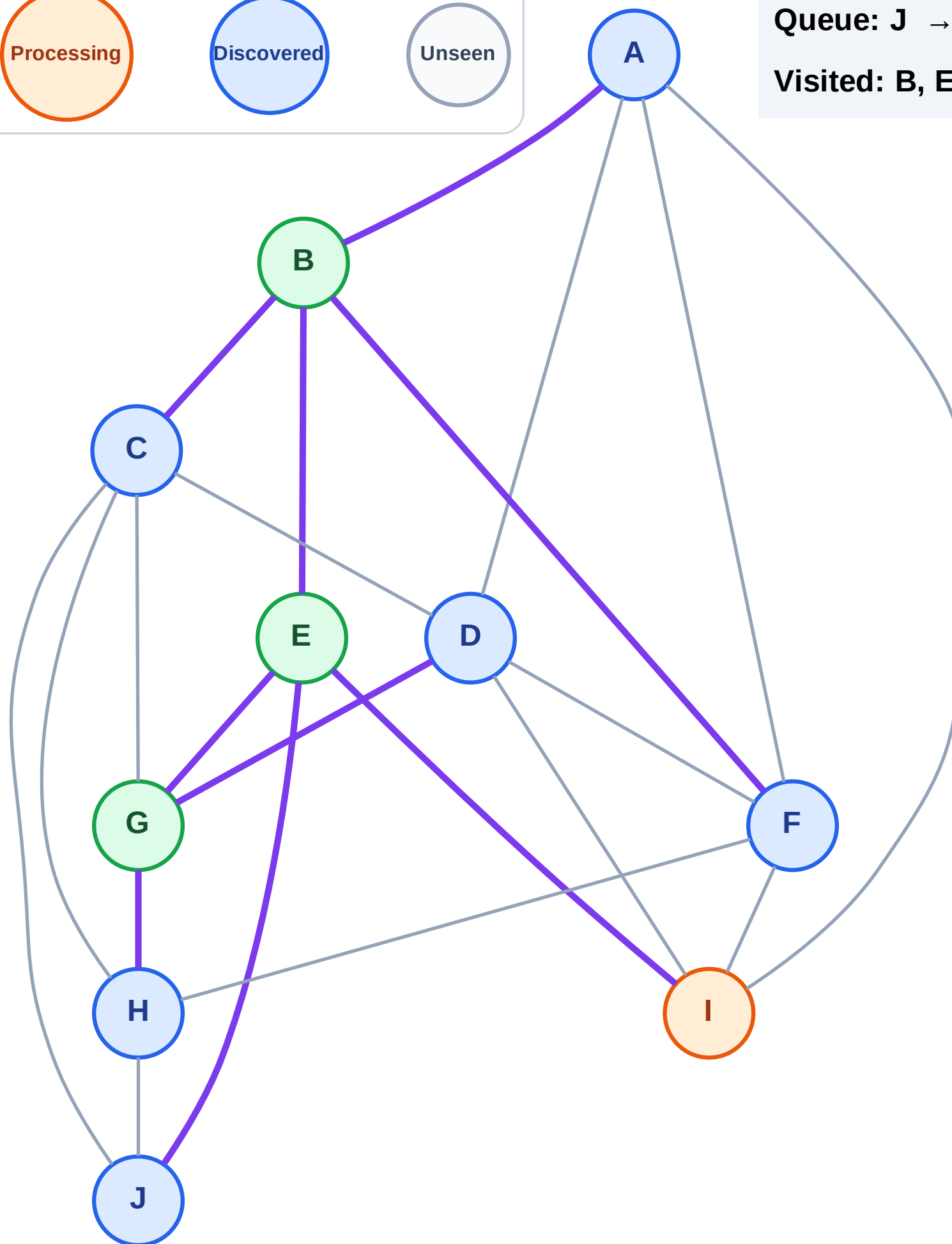
Discovered



Unseen

Queue: J → A → C → F → D → H

Visited: B, E, G



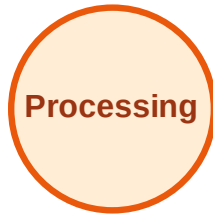
BFS Traversal

Step 9: No new neighbors from I

Legend



Complete



Processing



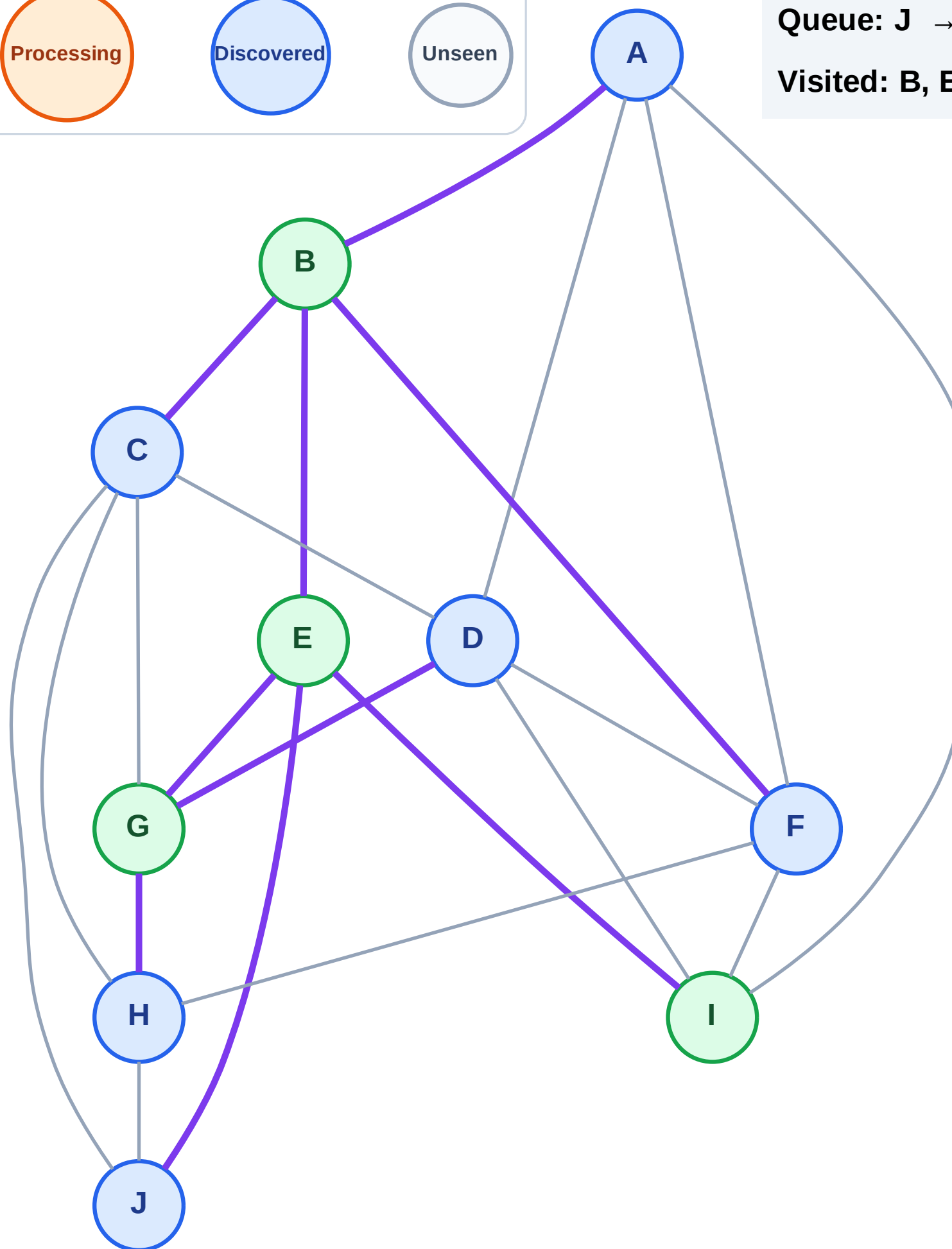
Discovered



Unseen

Queue: J → A → C → F → D → H

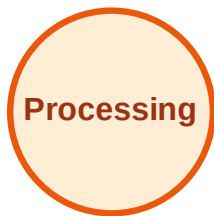
Visited: B, E, G, I



BFS Traversal

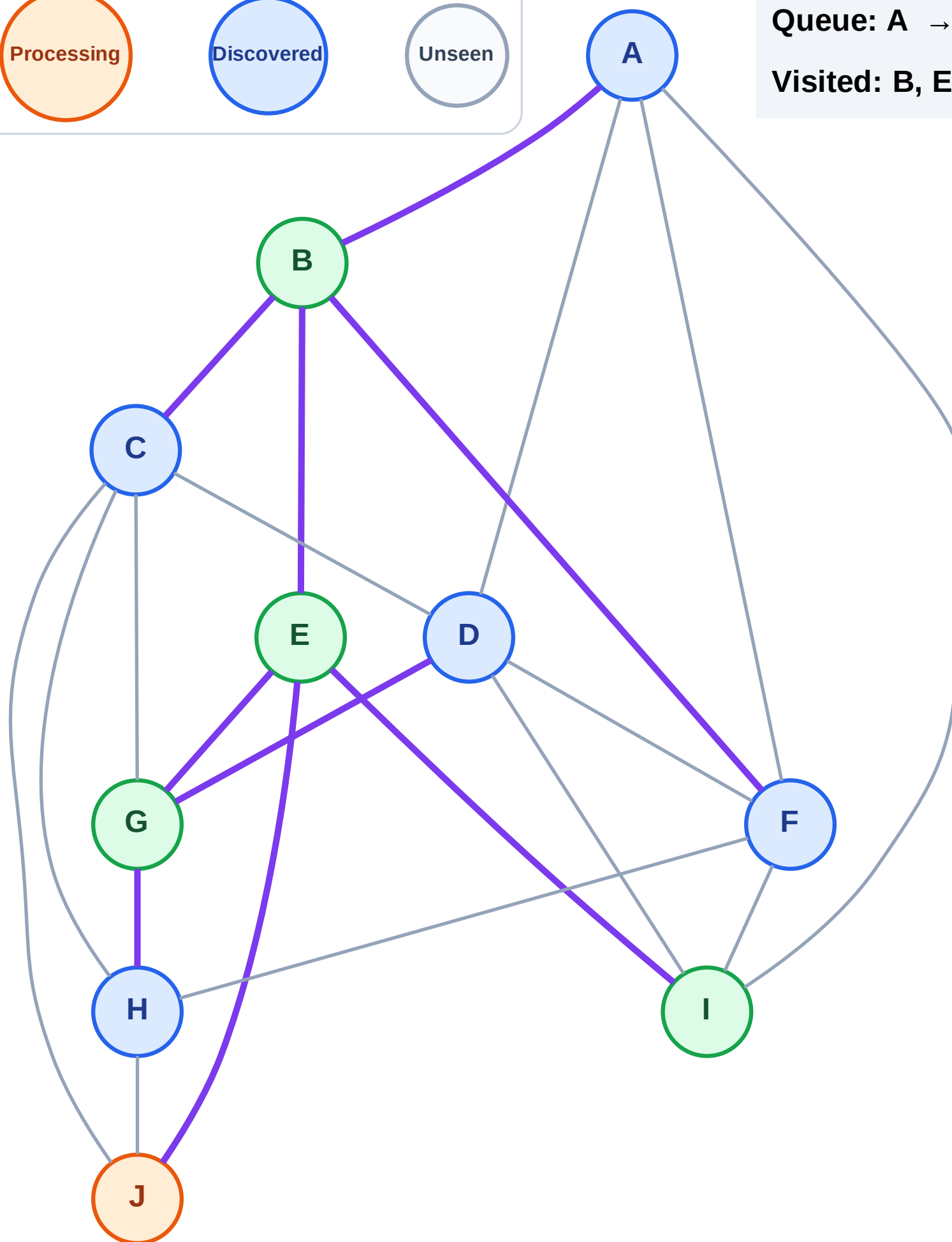
Step 10: Process node J

Legend



Queue: A → C → F → D → H

Visited: B, E, G, I



BFS Traversal

Step 11: No new neighbors from J

Legend

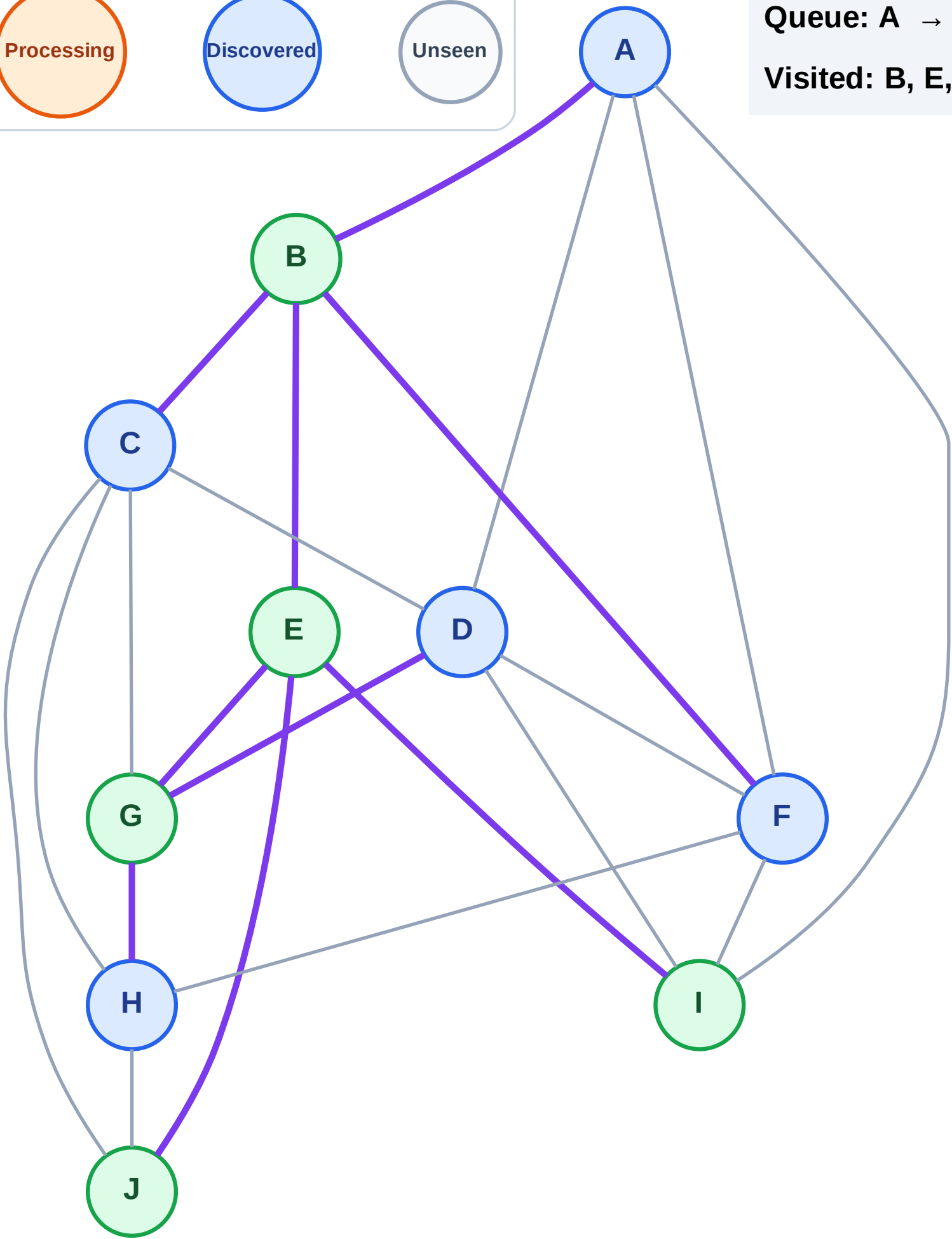


Processing



Queue: A → C → F → D → H

Visited: B, E, G, I, J



BFS Traversal

Step 12: Process node A

Legend

Complete

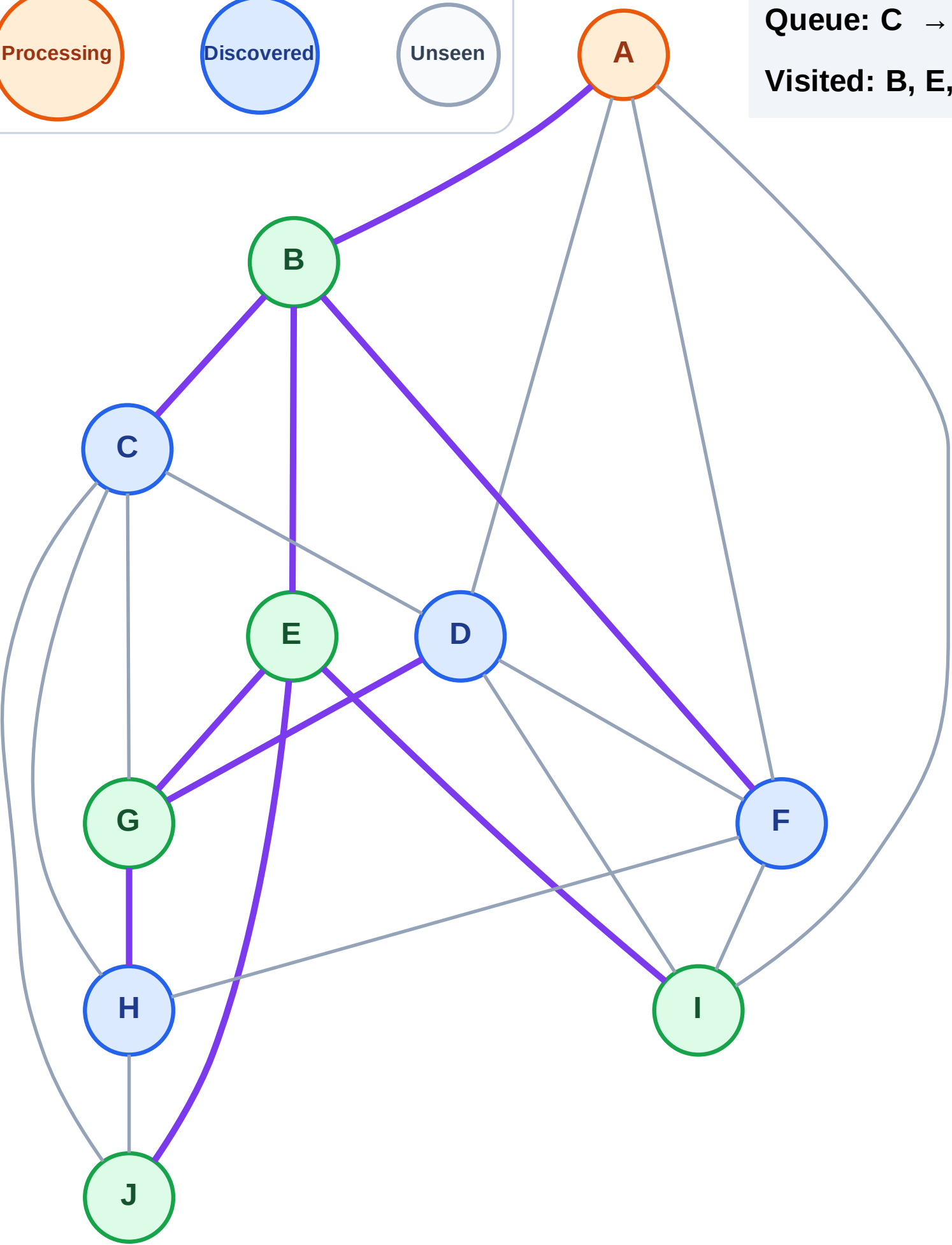
Processing

Discovered

Unseen

Queue: C → F → D → H

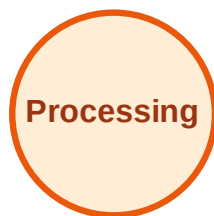
Visited: B, E, G, I, J



BFS Traversal

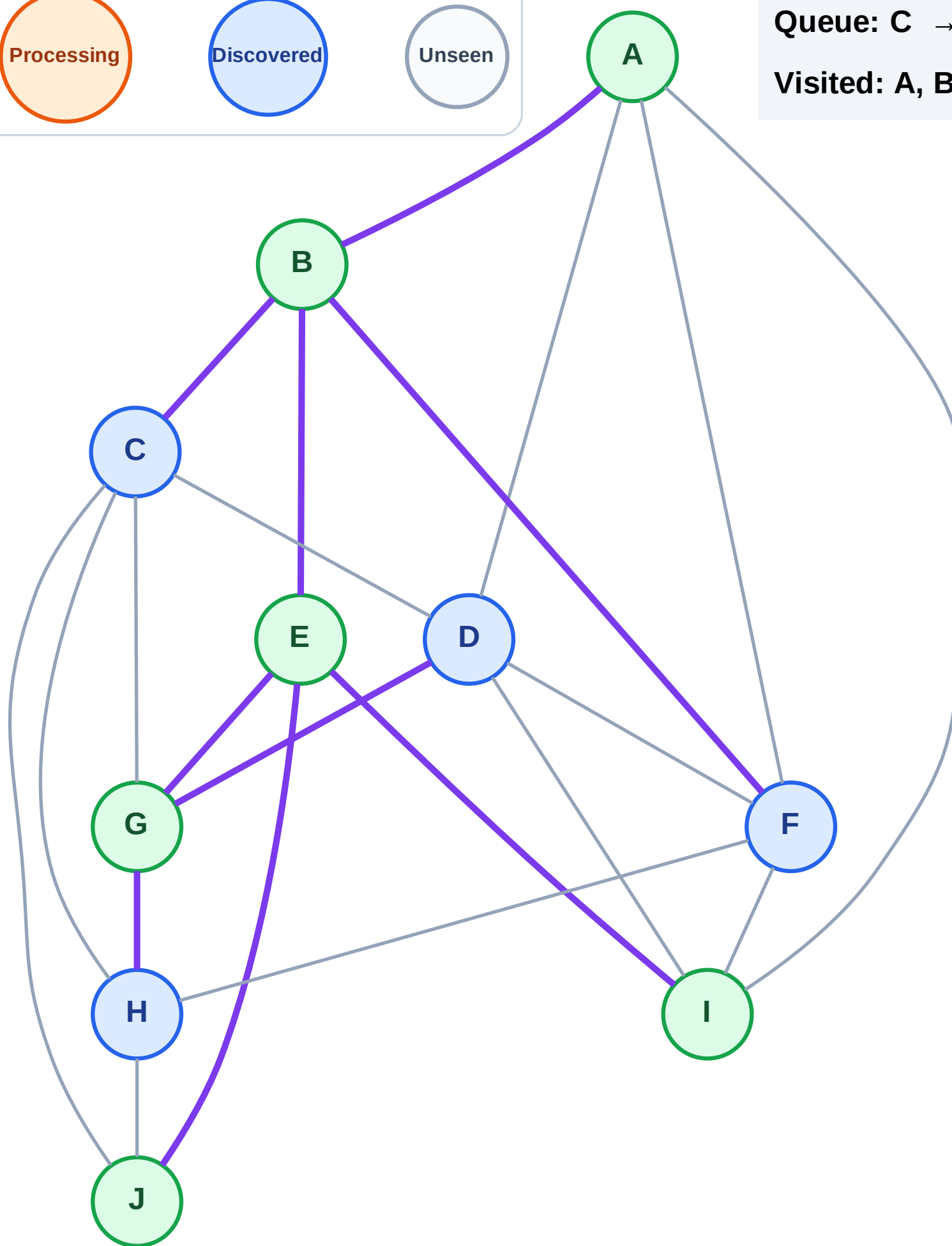
Step 13: No new neighbors from A

Legend



Queue: C → F → D → H

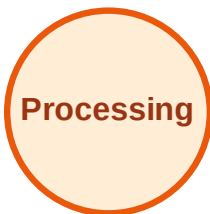
Visited: A, B, E, G, I, J



BFS Traversal

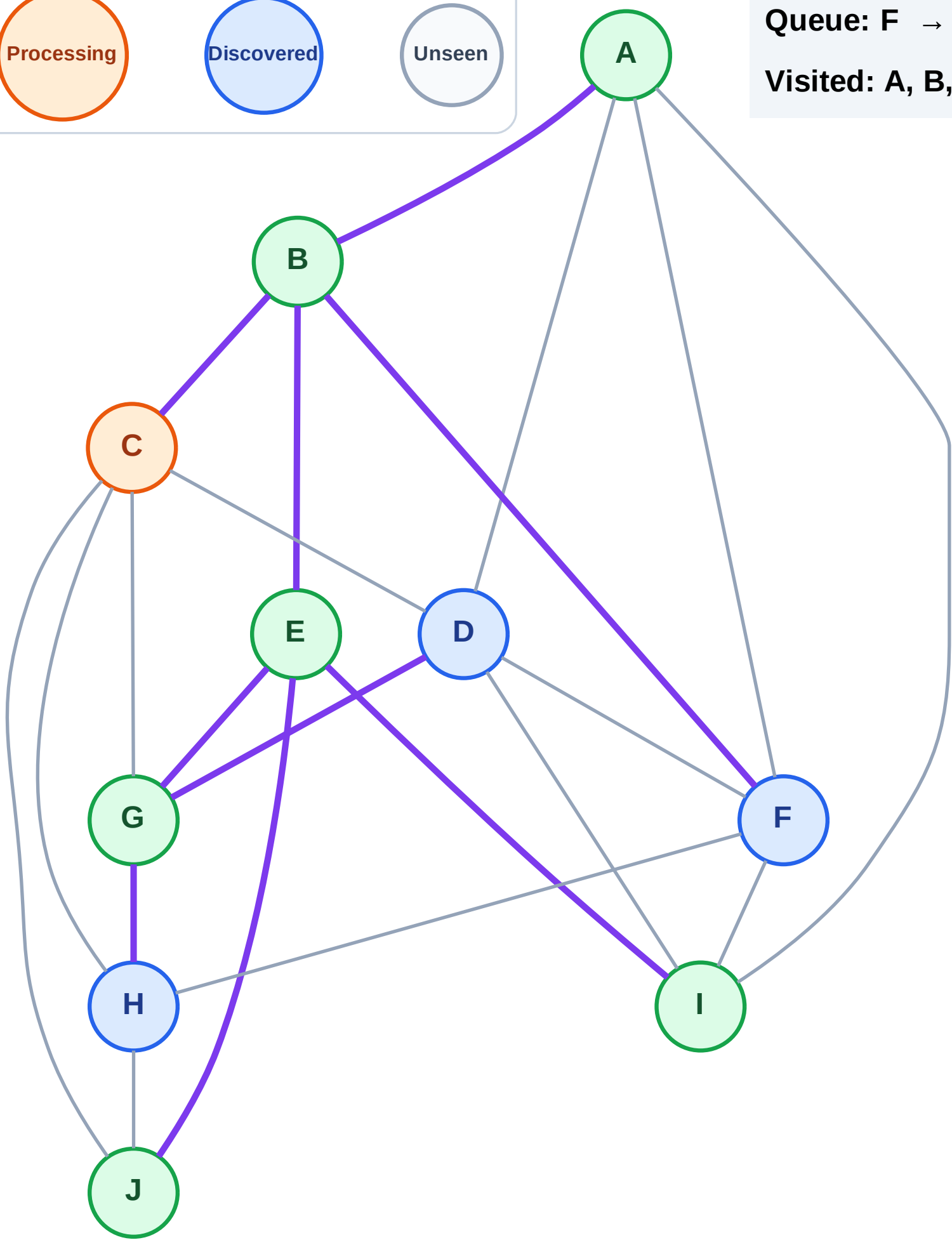
Step 14: Process node C

Legend



Queue: F → D → H

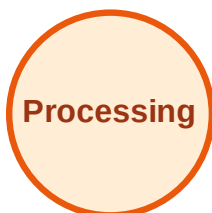
Visited: A, B, E, G, I, J



BFS Traversal

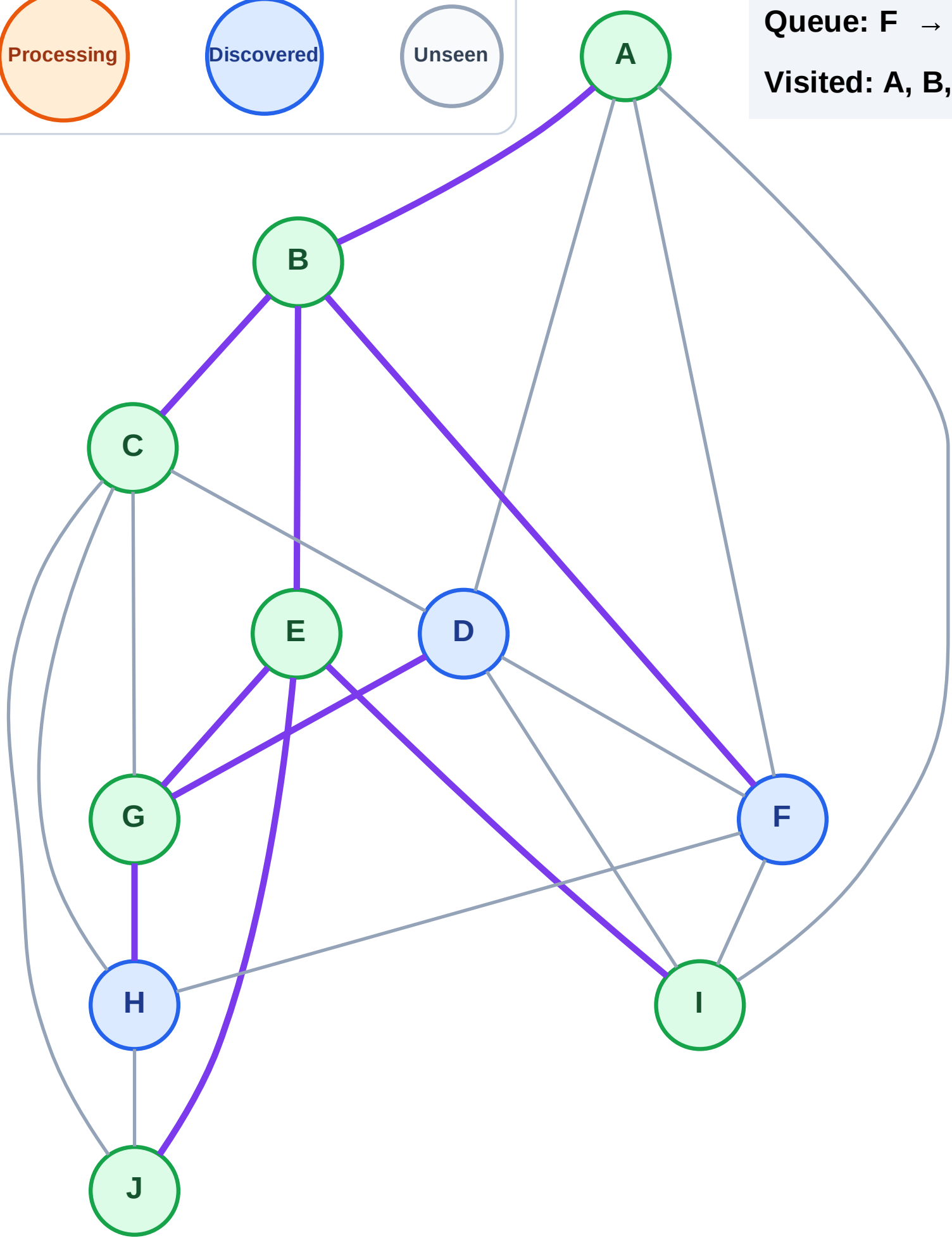
Step 15: No new neighbors from C

Legend



Queue: F → D → H

Visited: A, B, C, E, G, I, J



BFS Traversal

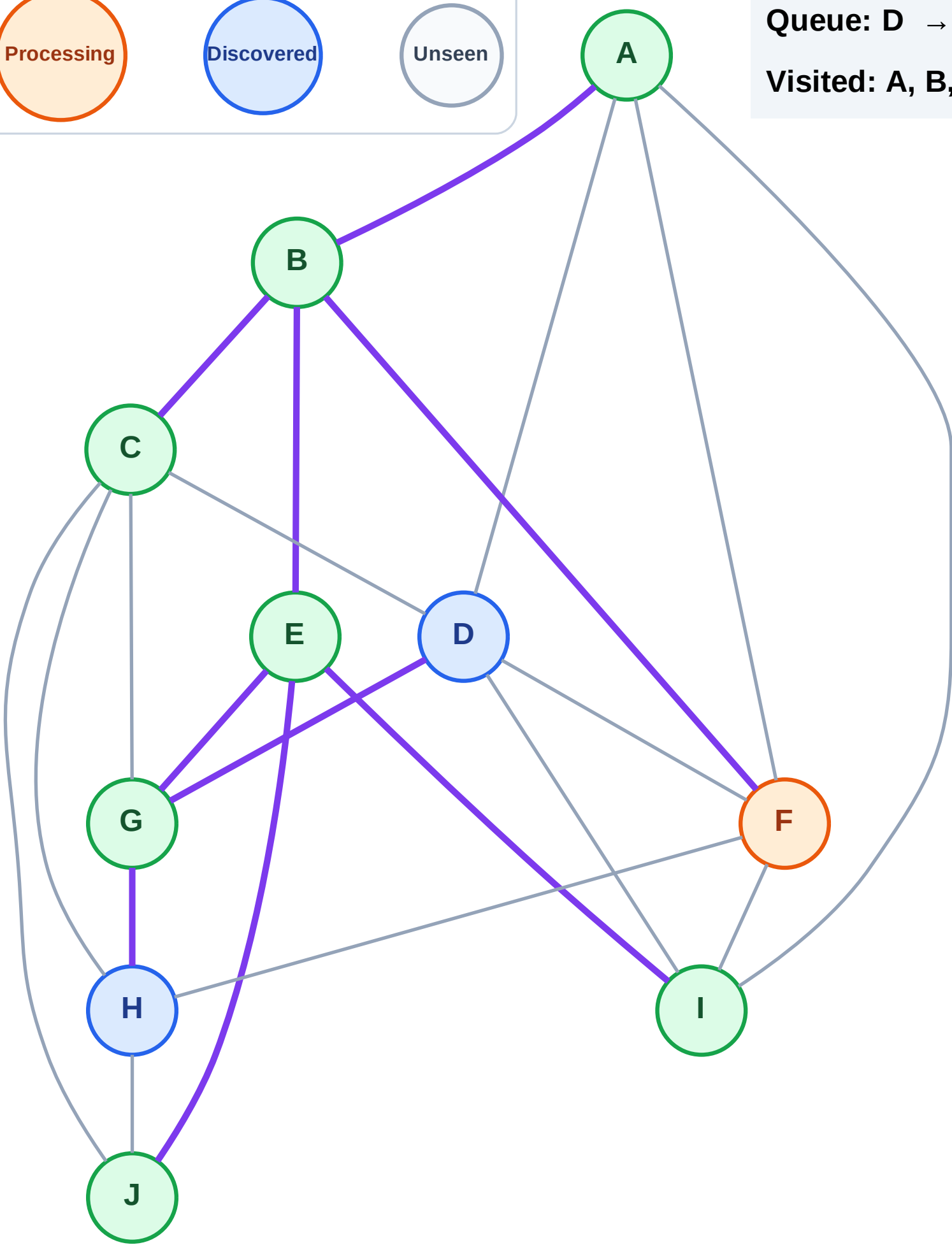
Step 16: Process node F

Legend



Queue: D → H

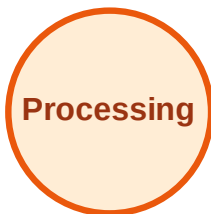
Visited: A, B, C, E, G, I, J



BFS Traversal

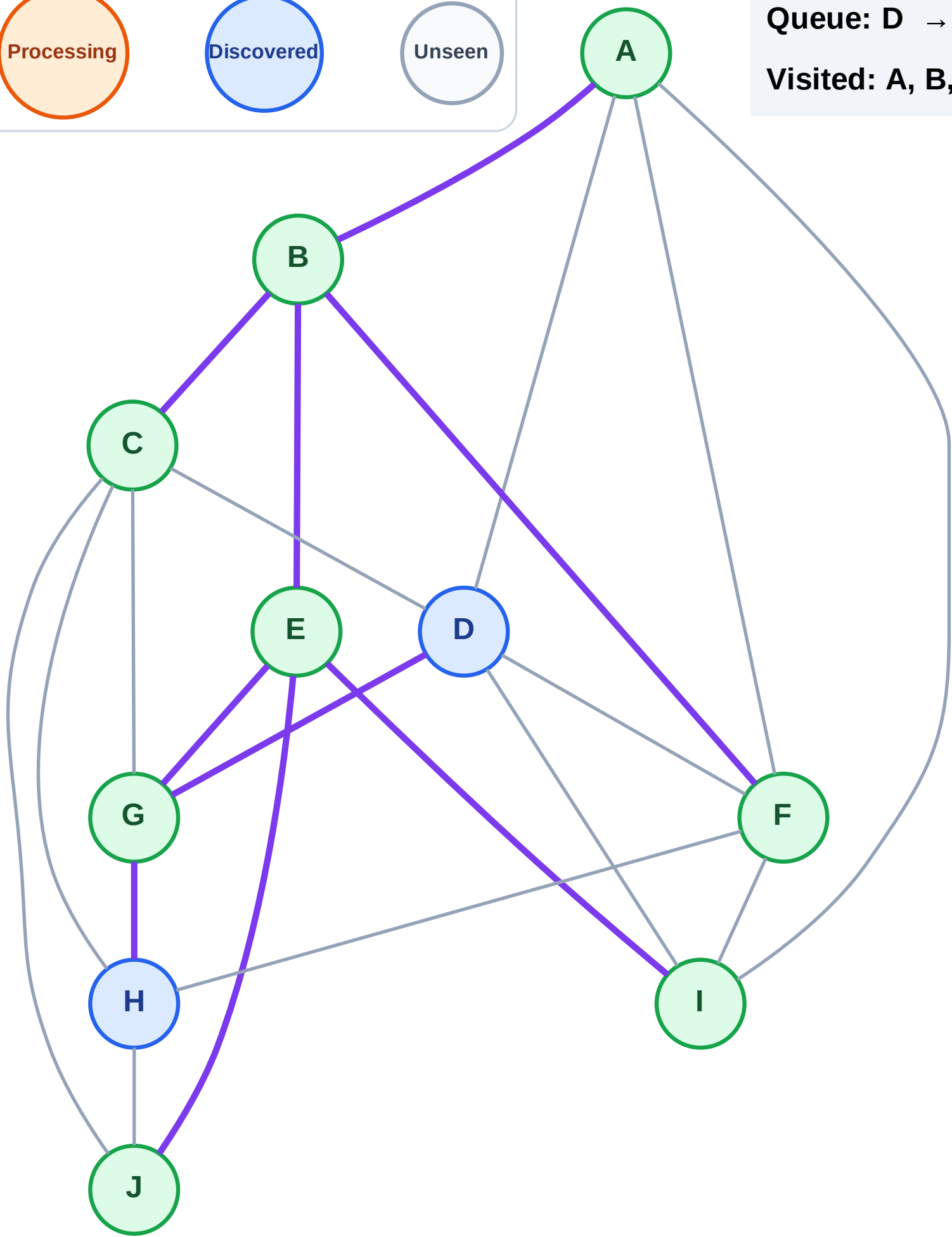
Step 17: No new neighbors from F

Legend



Queue: D → H

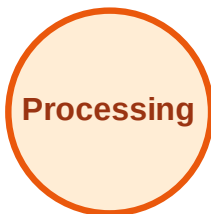
Visited: A, B, C, E, F, G, I, J



BFS Traversal

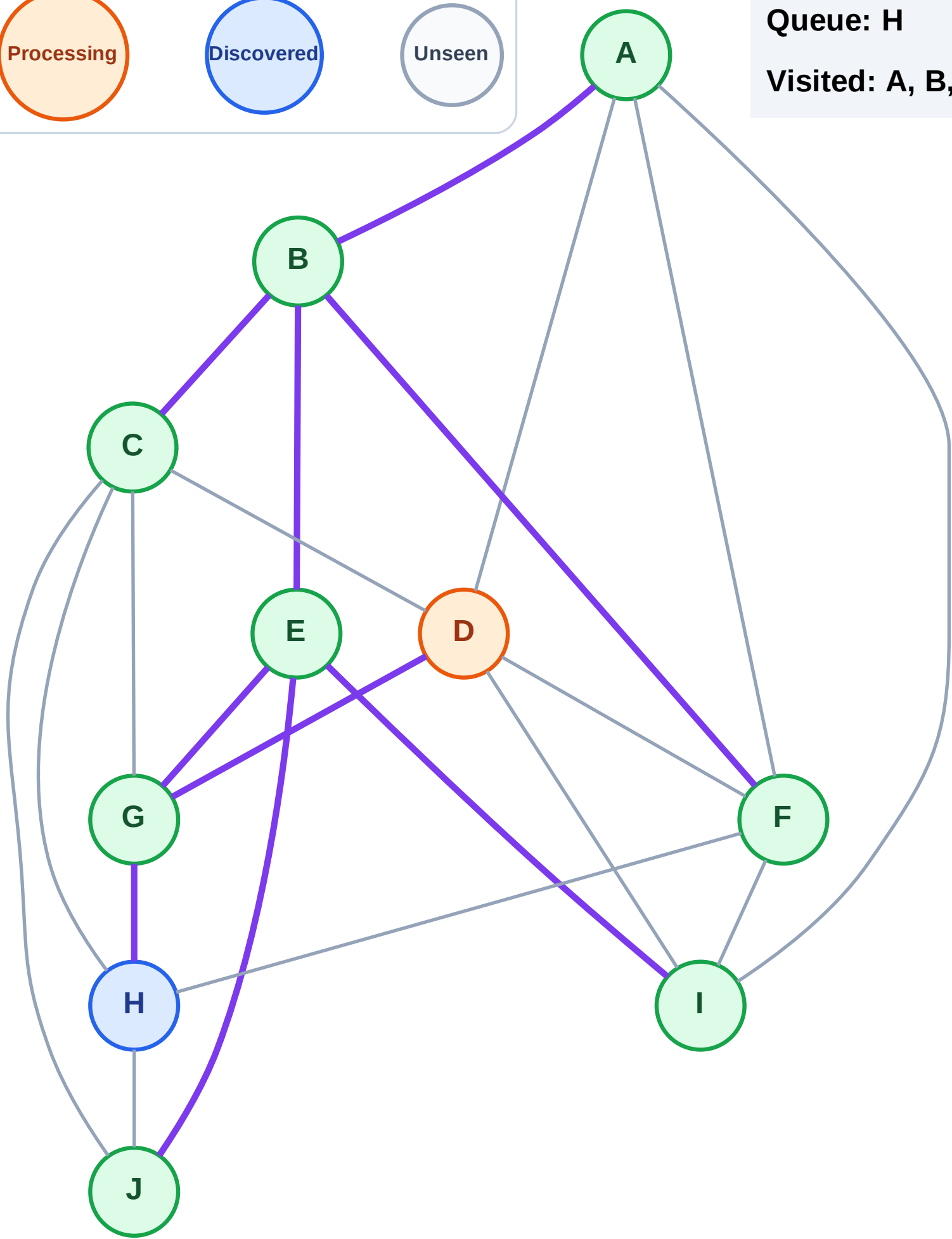
Step 18: Process node D

Legend



Queue: H

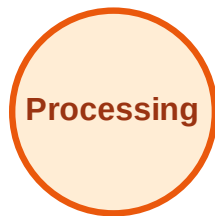
Visited: A, B, C, E, F, G, I, J



BFS Traversal

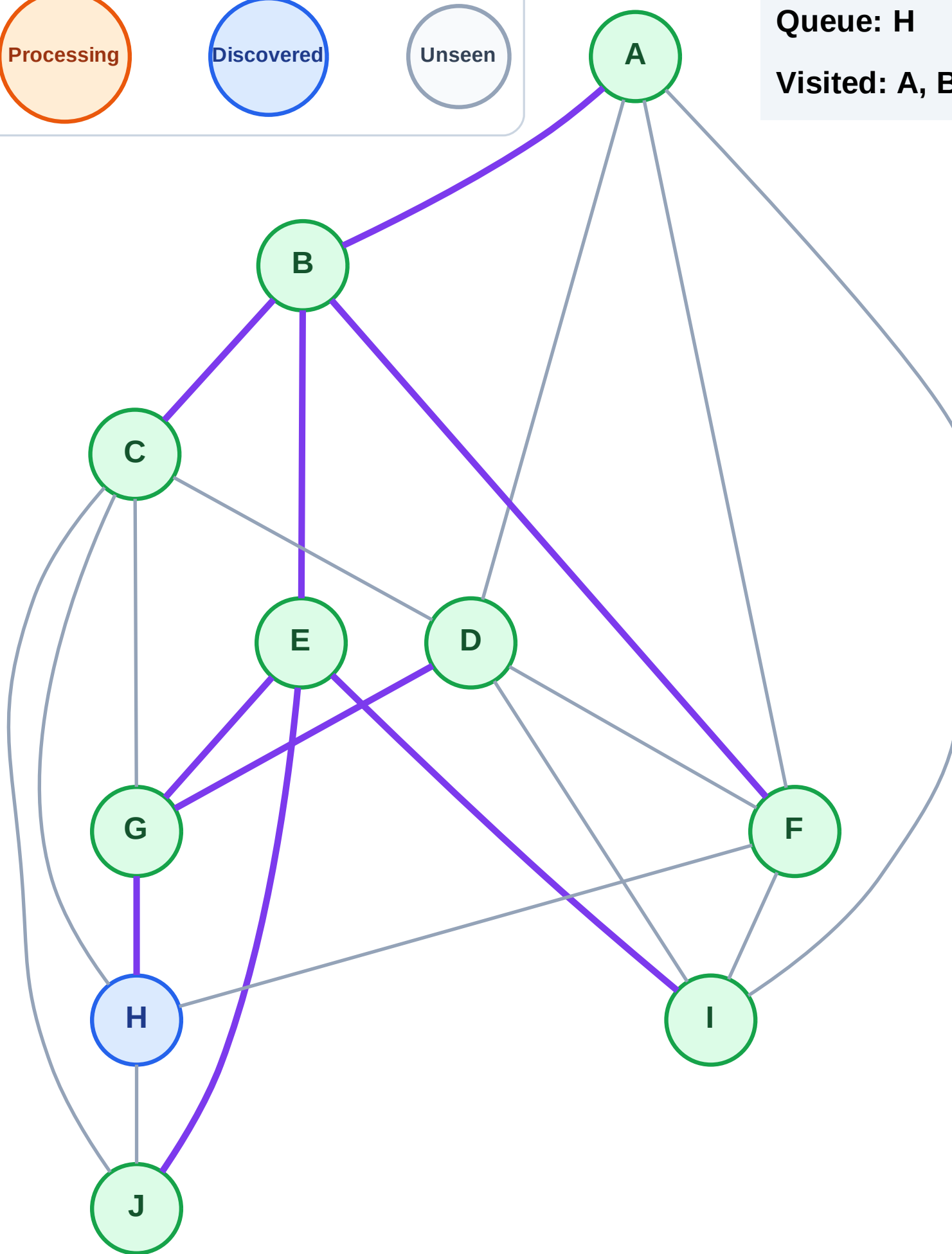
Step 19: No new neighbors from D

Legend



Queue: H

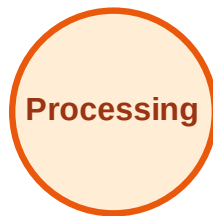
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

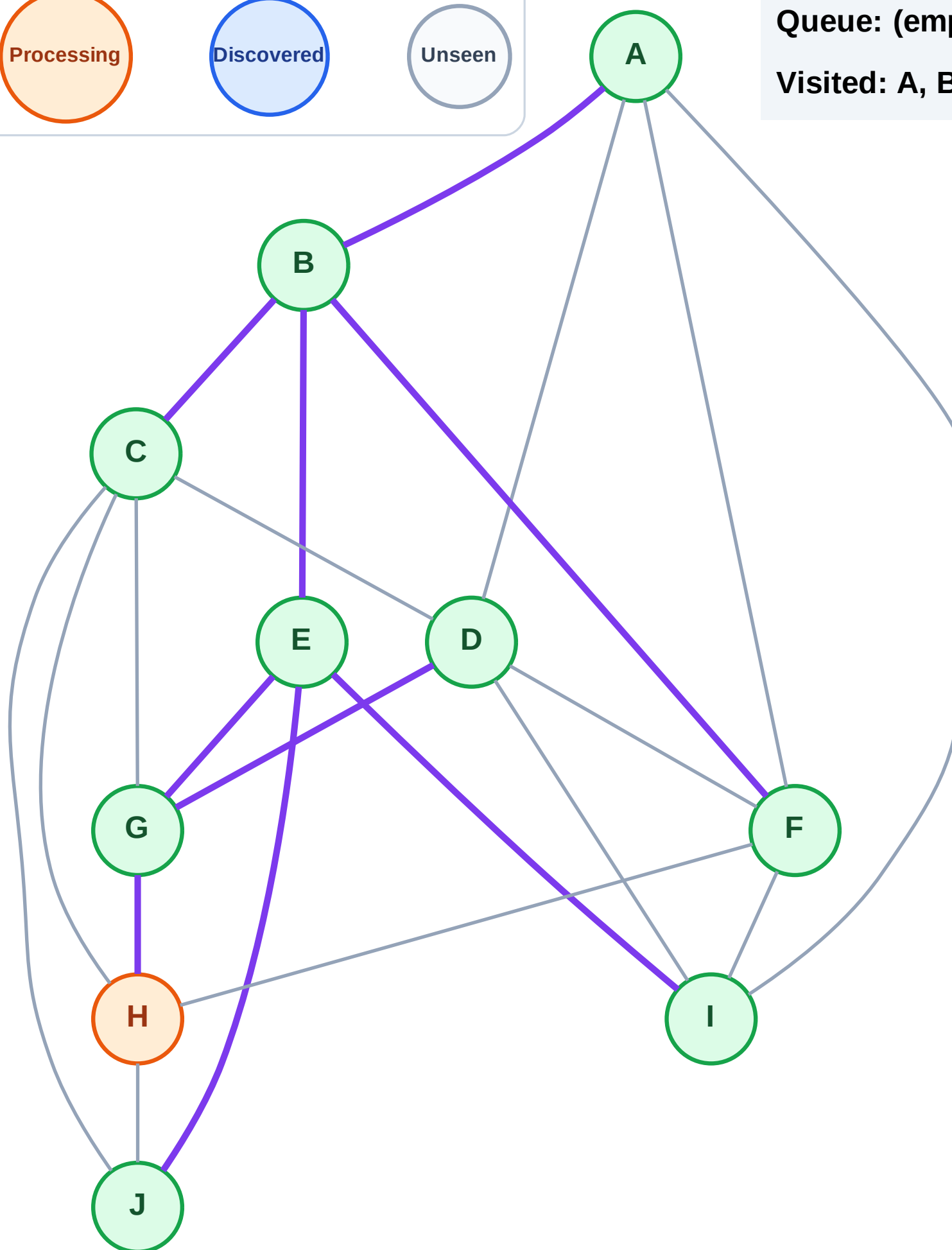
Step 20: Process node H

Legend



Queue: (empty)

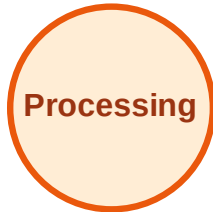
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

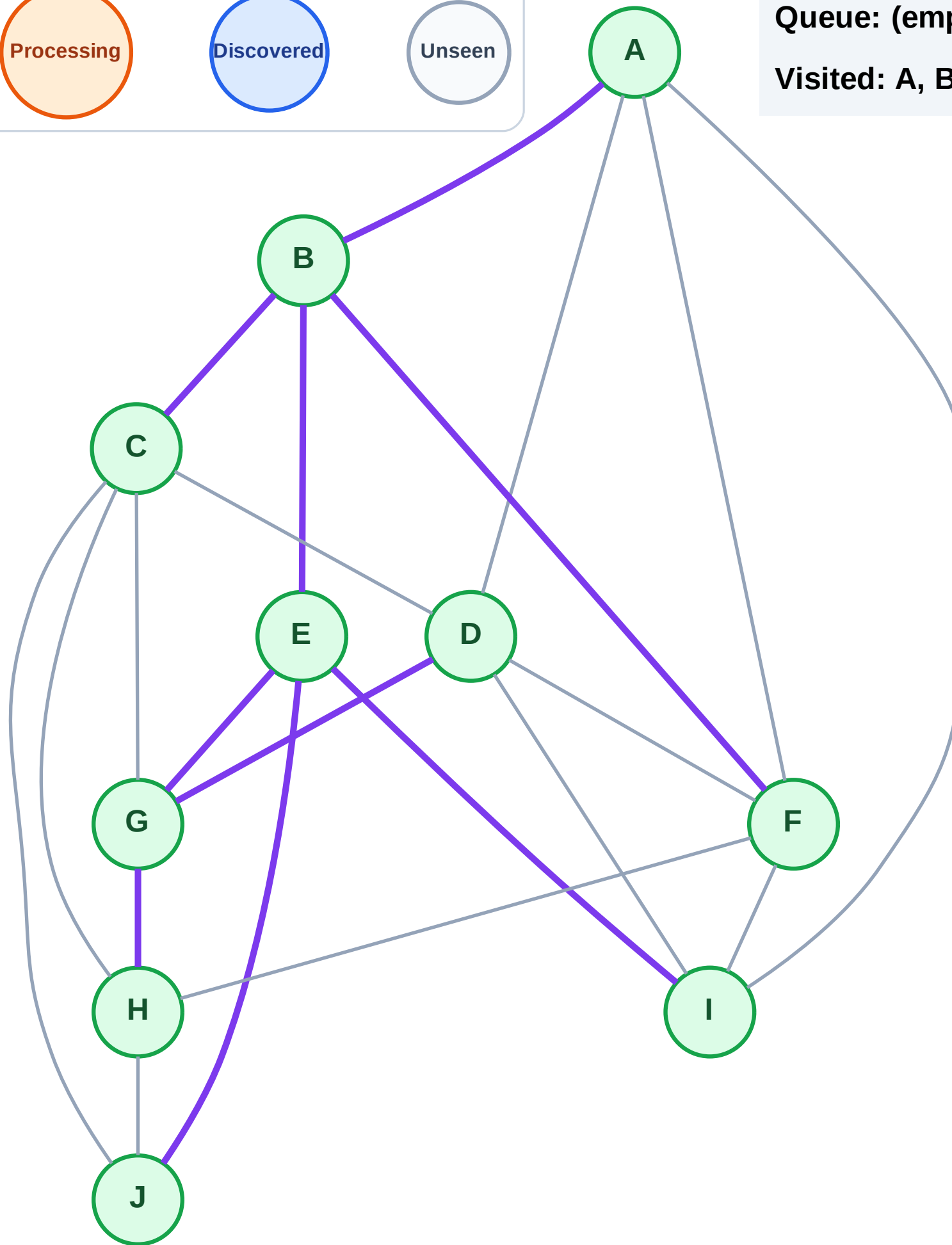
Step 21: No new neighbors from H

Legend



Queue: (empty)

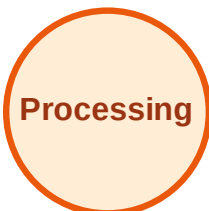
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

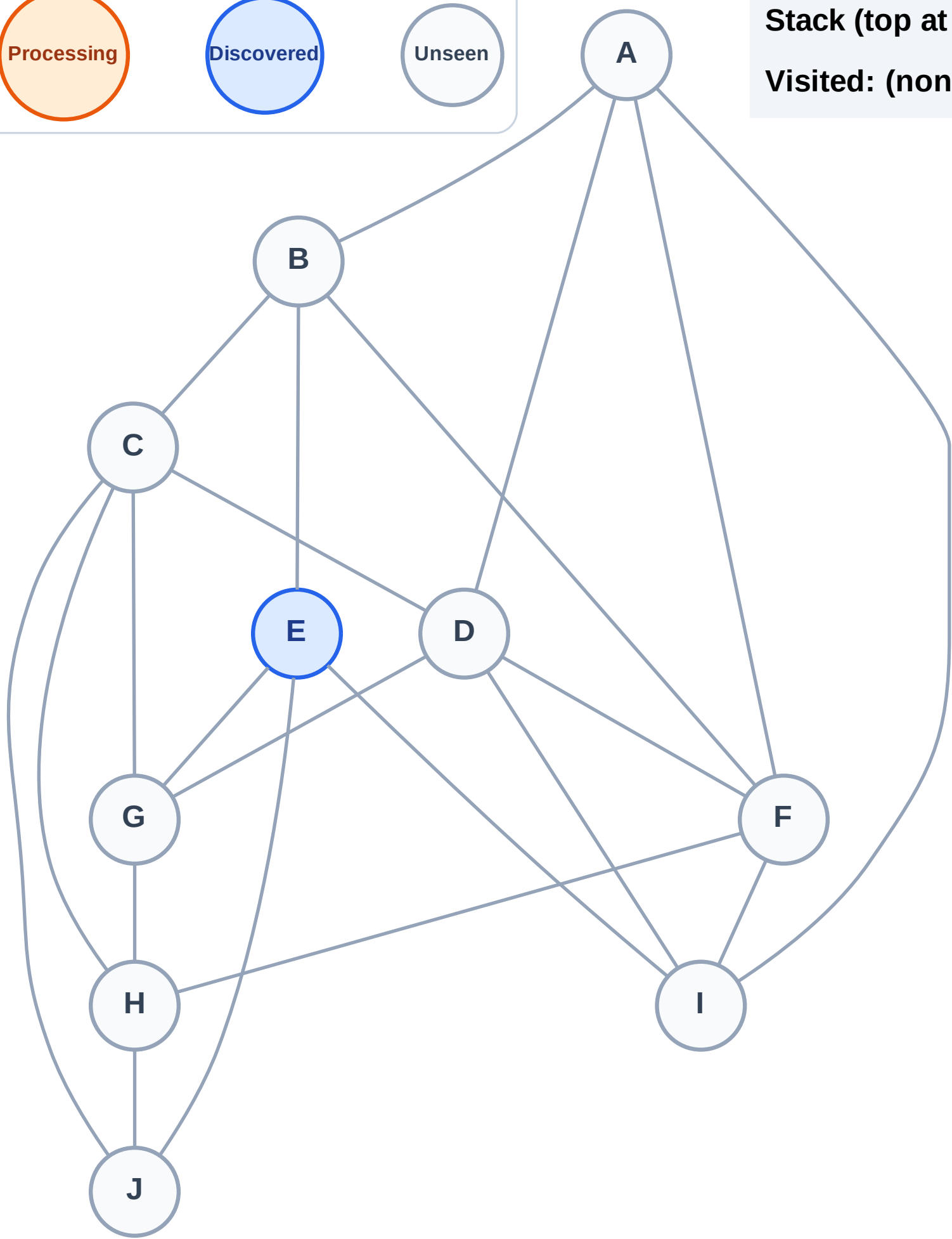
Step 1: Start at E

Legend



Stack (top at right): E

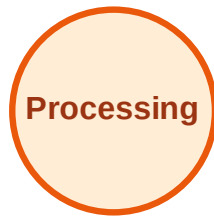
Visited: (none)



DFS Traversal

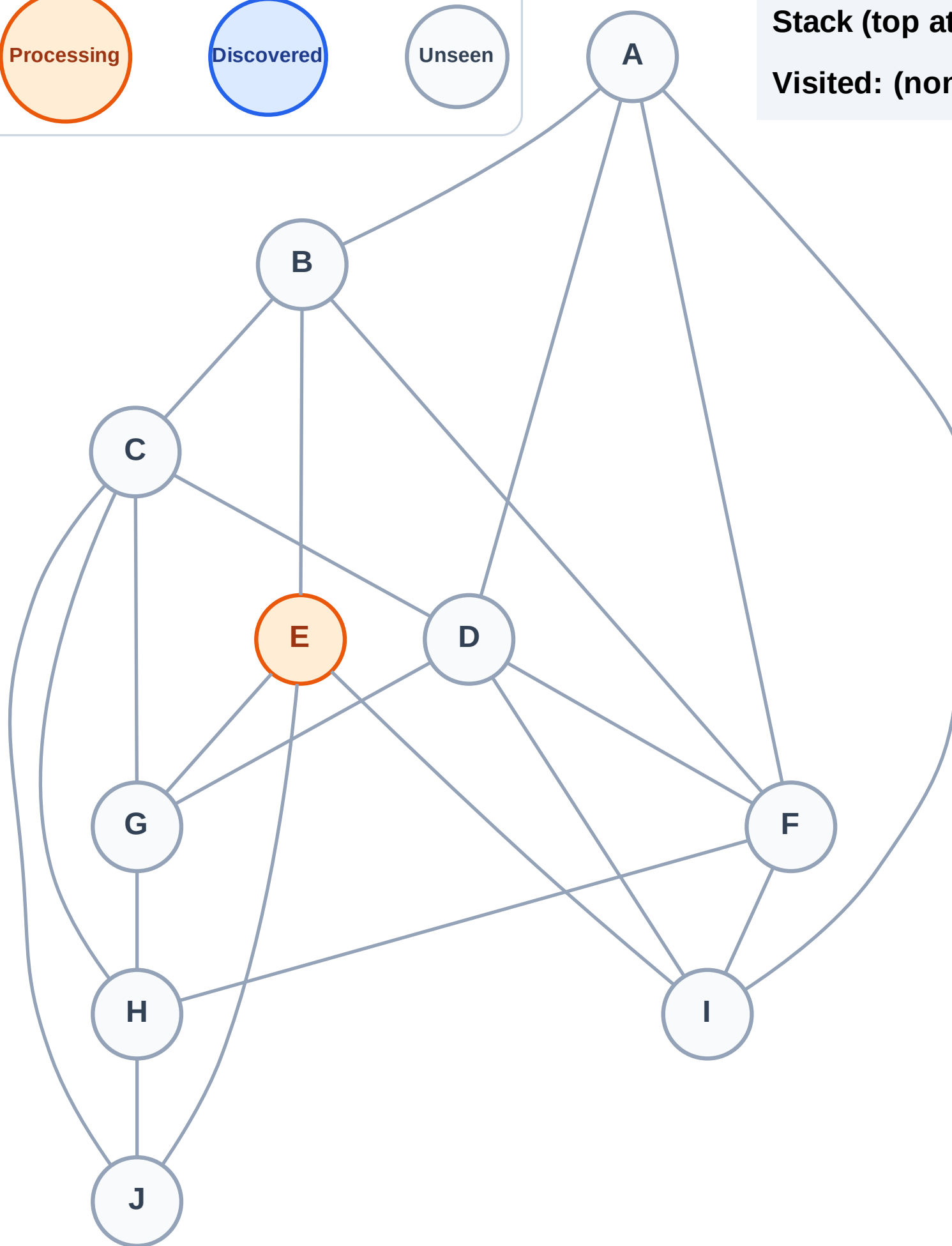
Step 2: Process node E

Legend



Stack (top at right): (empty)

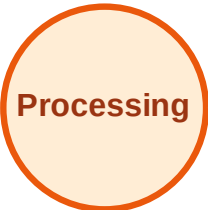
Visited: (none)



DFS Traversal

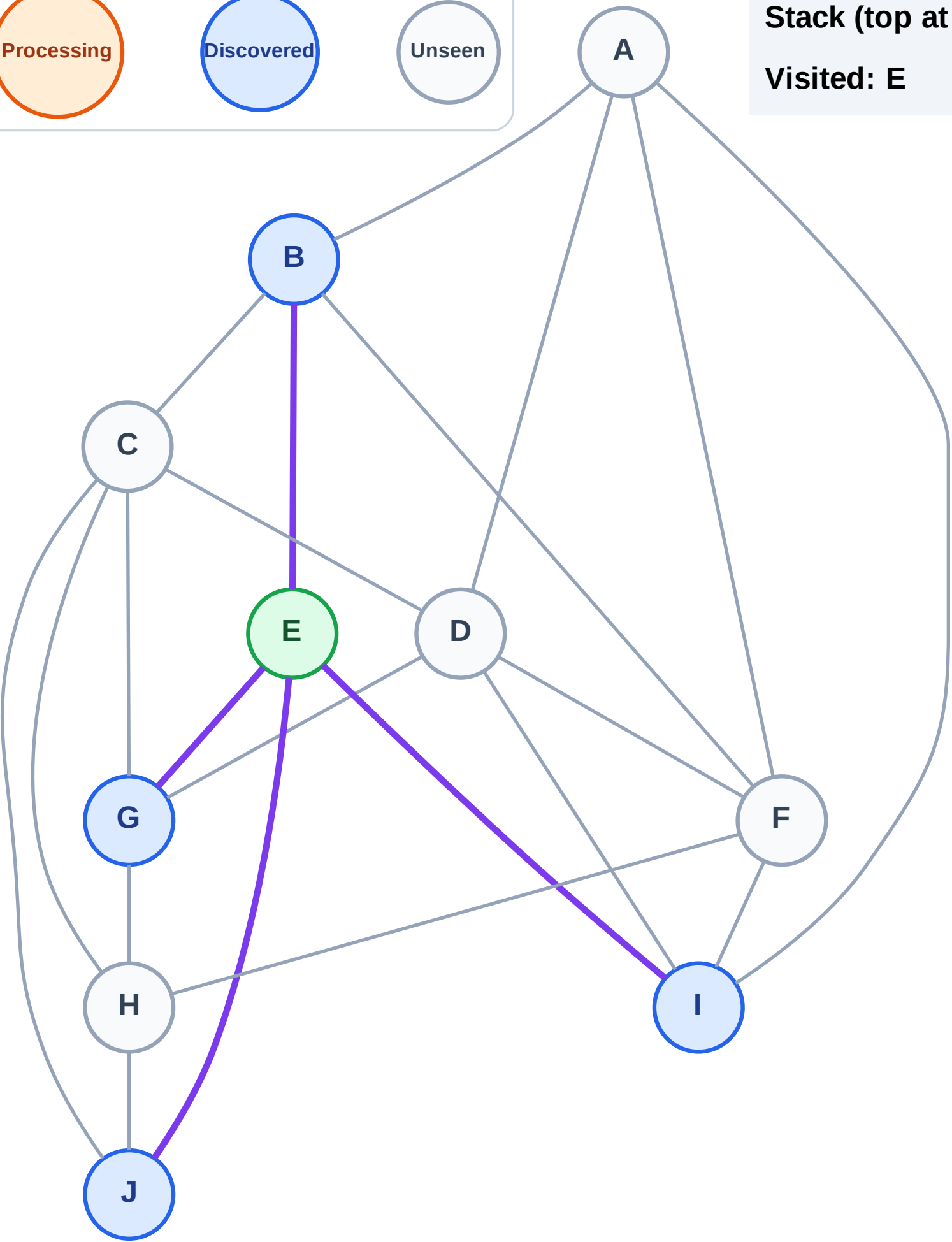
Step 3: Discover J, I, G, B from E

Legend



Stack (top at right): J | I | G | B

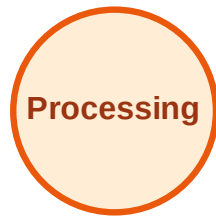
Visited: E



DFS Traversal

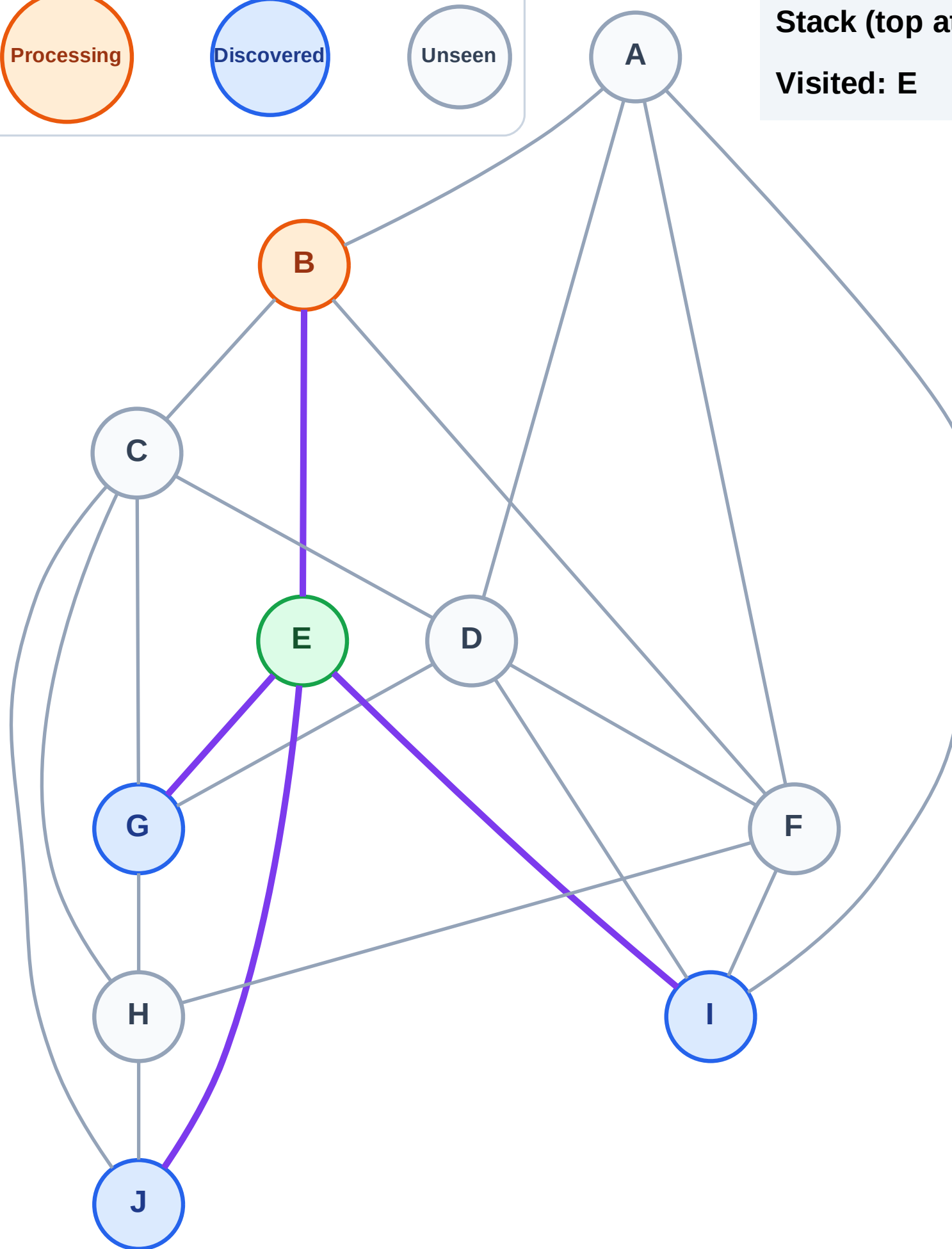
Step 4: Process node B

Legend



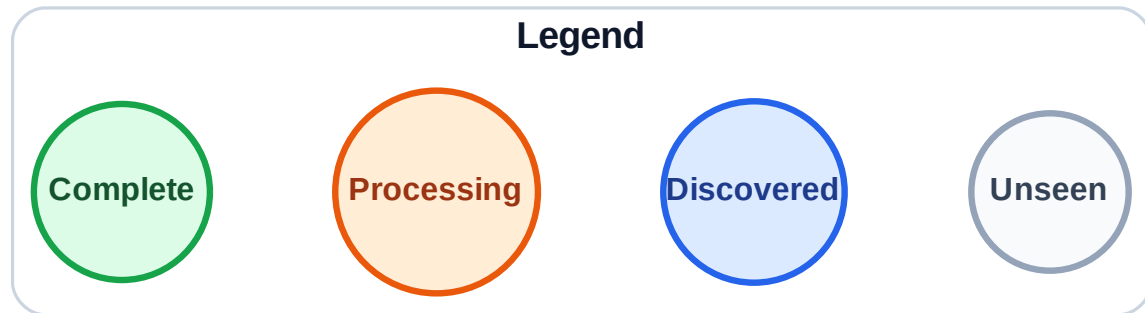
Stack (top at right): J | I | G

Visited: E



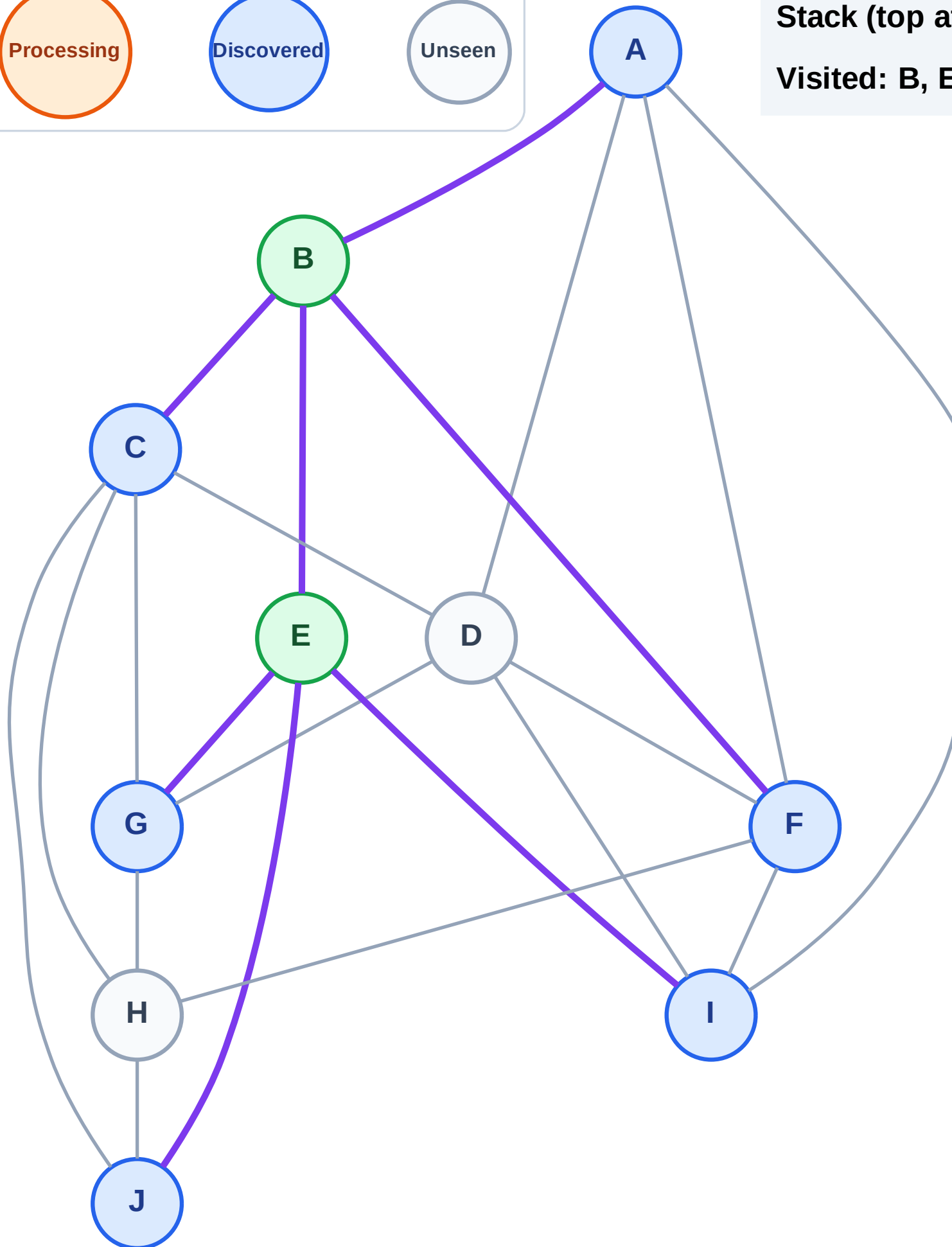
Step 5: Discover F, C, A from B

Step 5: Discover F, C, A from B



Stack (top at right): J | I | G | F | C | A

Visited: B, E



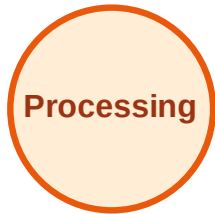
DFS Traversal

Step 6: Process node A

Legend



Complete



Processing



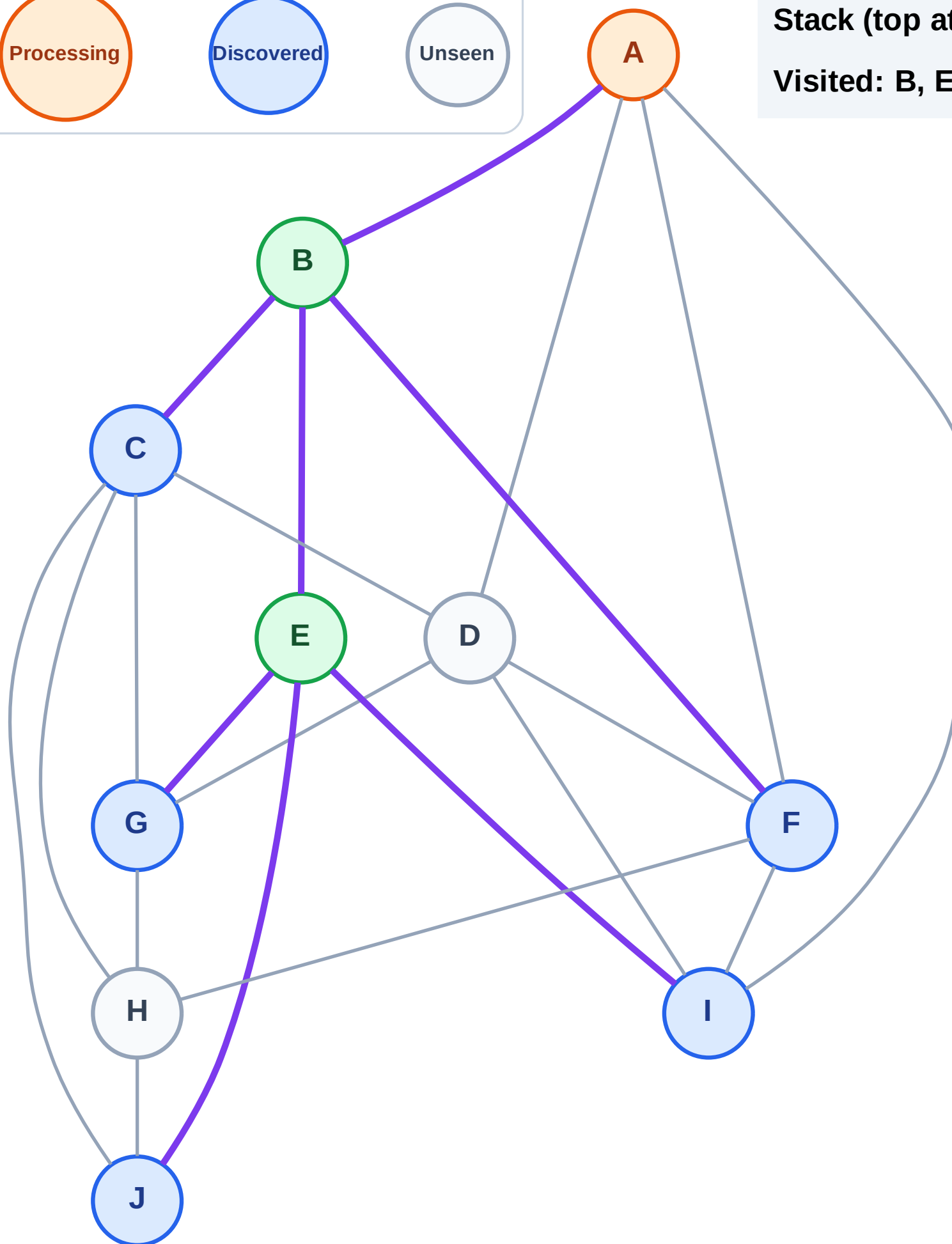
Discovered



Unseen

Stack (top at right): J | I | G | F | C

Visited: B, E



DFS Traversal

Step 7: Discover D from A

Legend

Complete

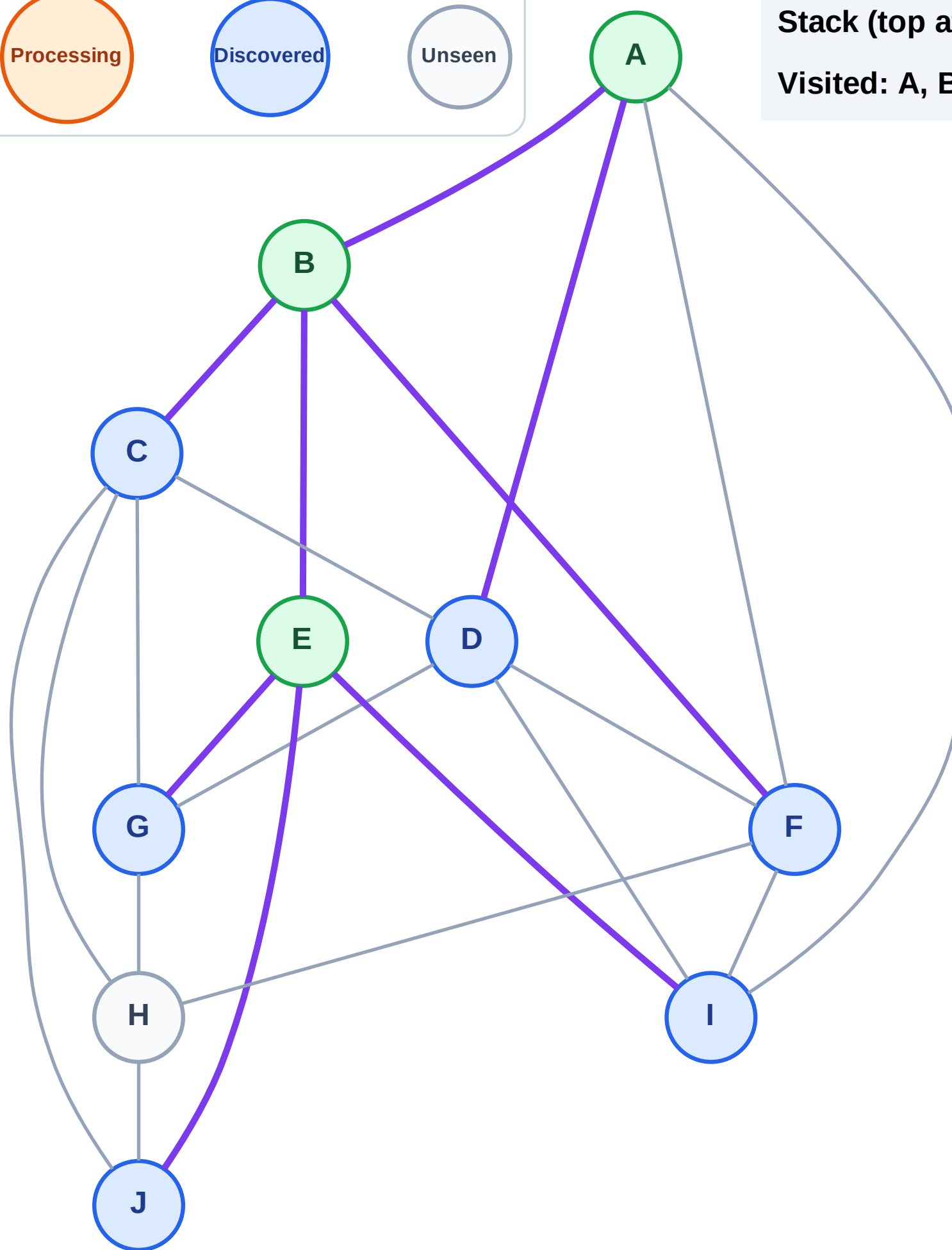
Processing

Discovered

Unseen

Stack (top at right): J | I | G | F | C | D

Visited: A, B, E



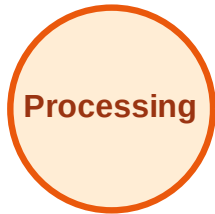
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



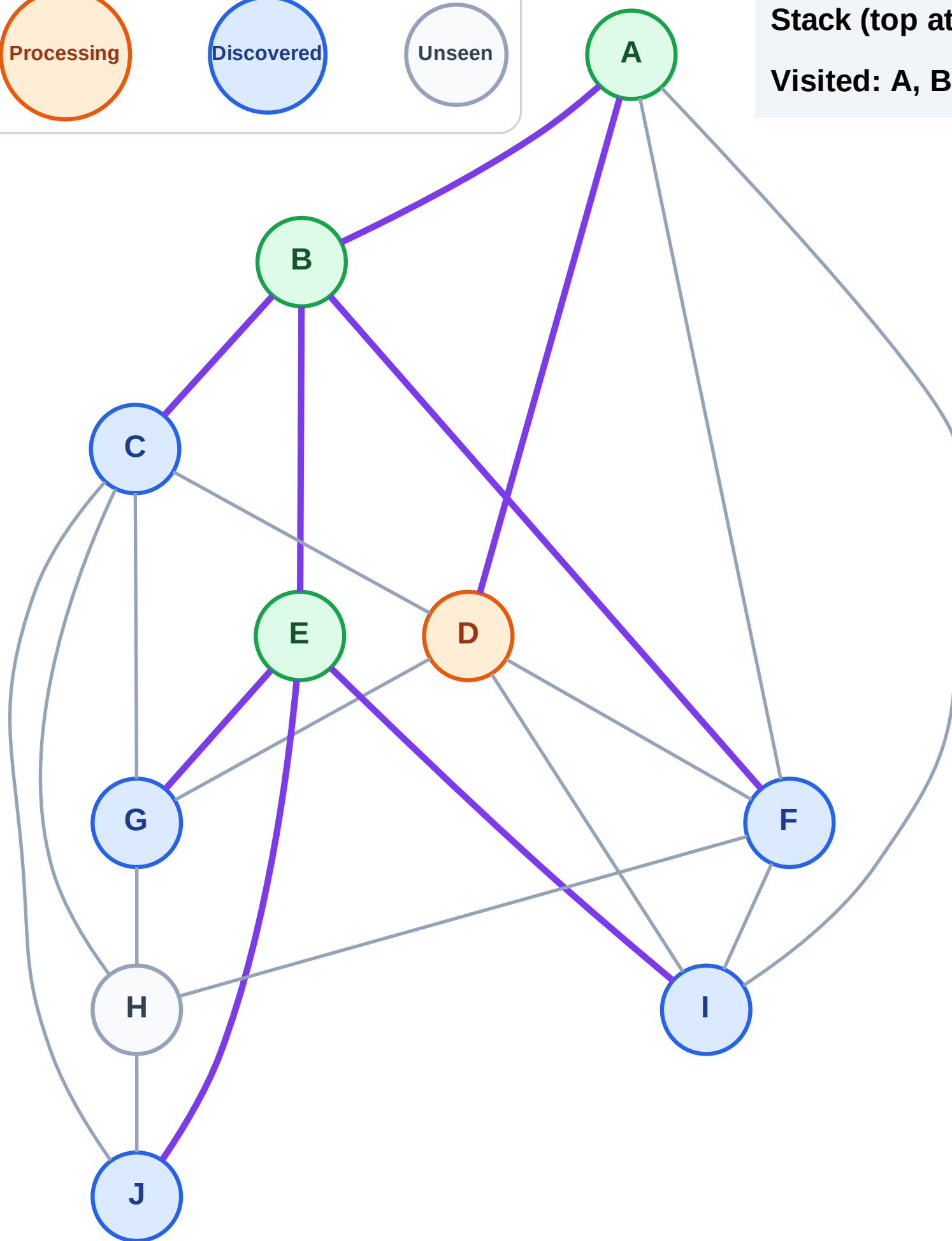
Discovered



Unseen

Stack (top at right): J | I | G | F | C

Visited: A, B, E



DFS Traversal

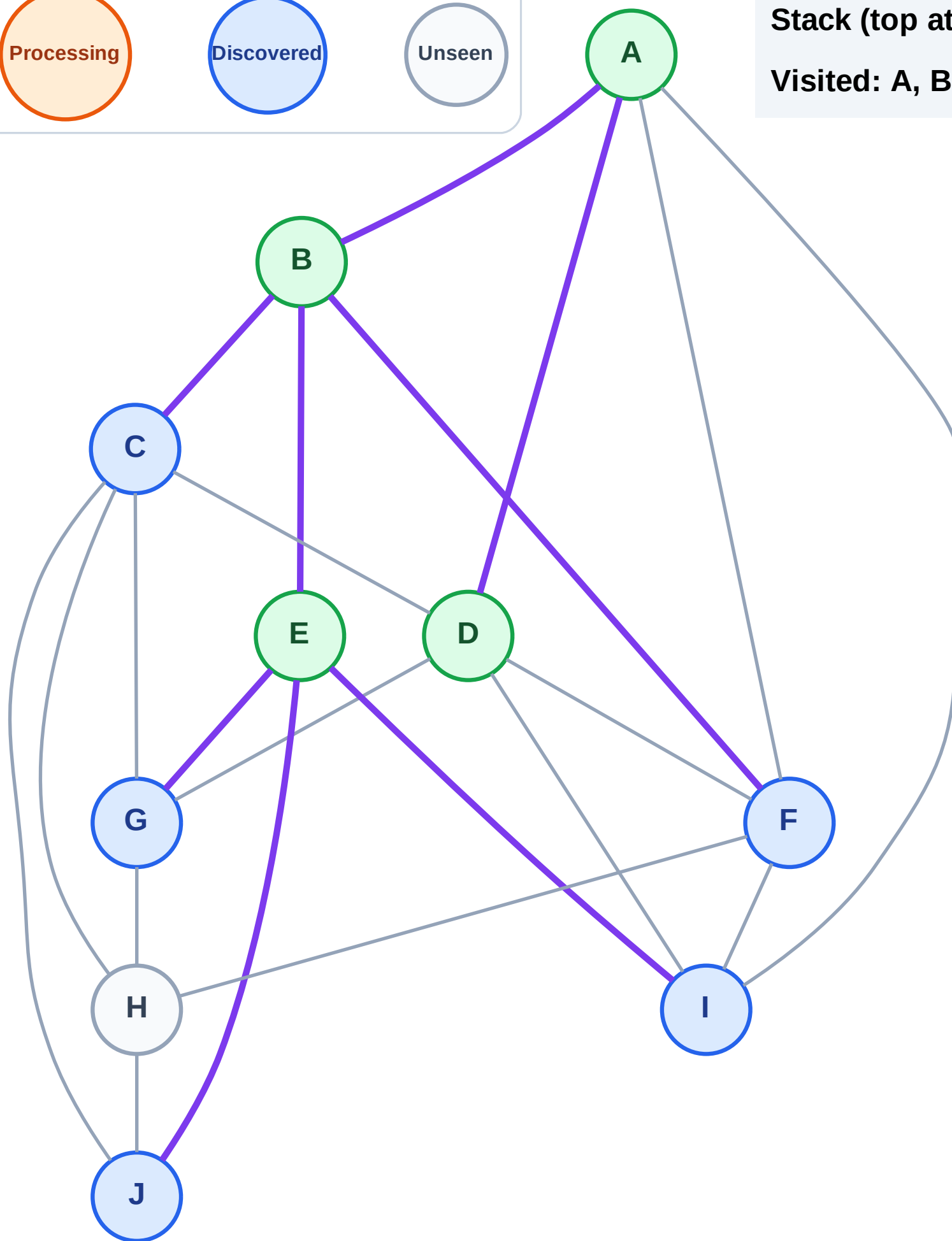
Step 9: No new neighbors from D

Legend



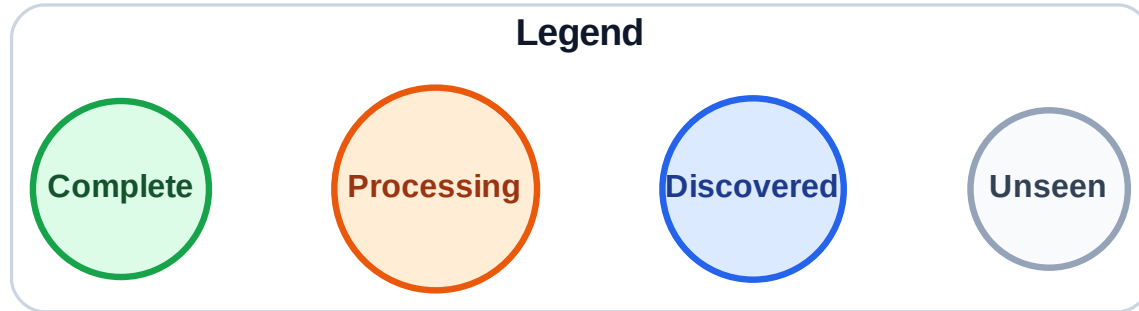
Stack (top at right): J | I | G | F | C

Visited: A, B, D, E



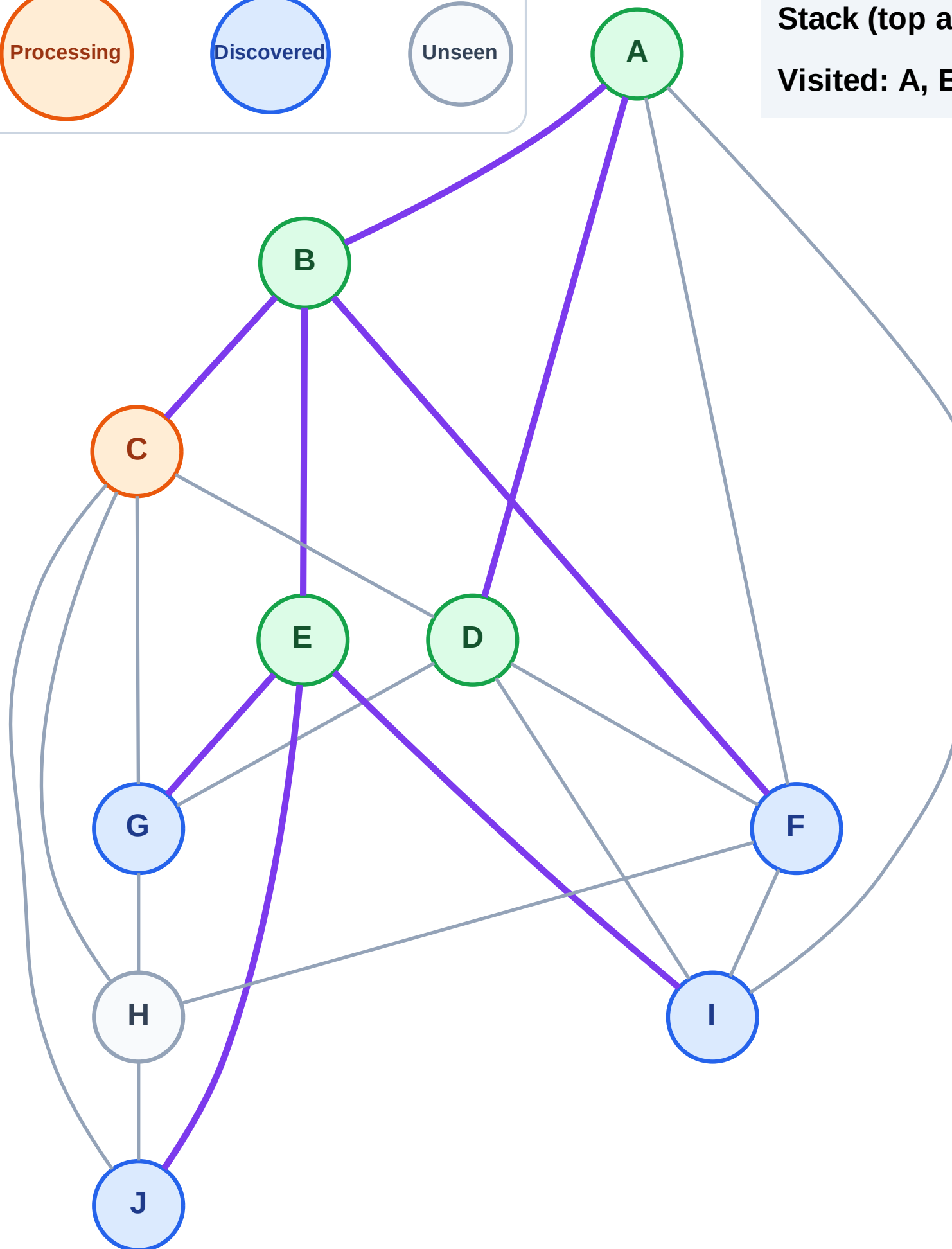
DFS Traversal

Step 10: Process node C



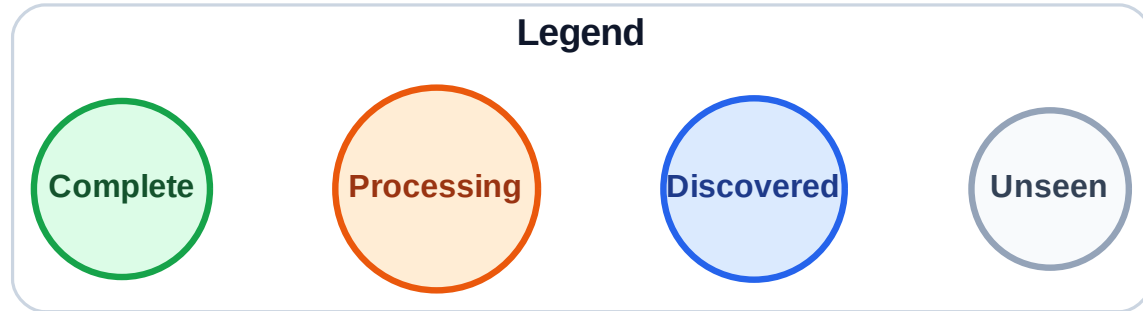
Stack (top at right): J | I | G | F

Visited: A, B, D, E



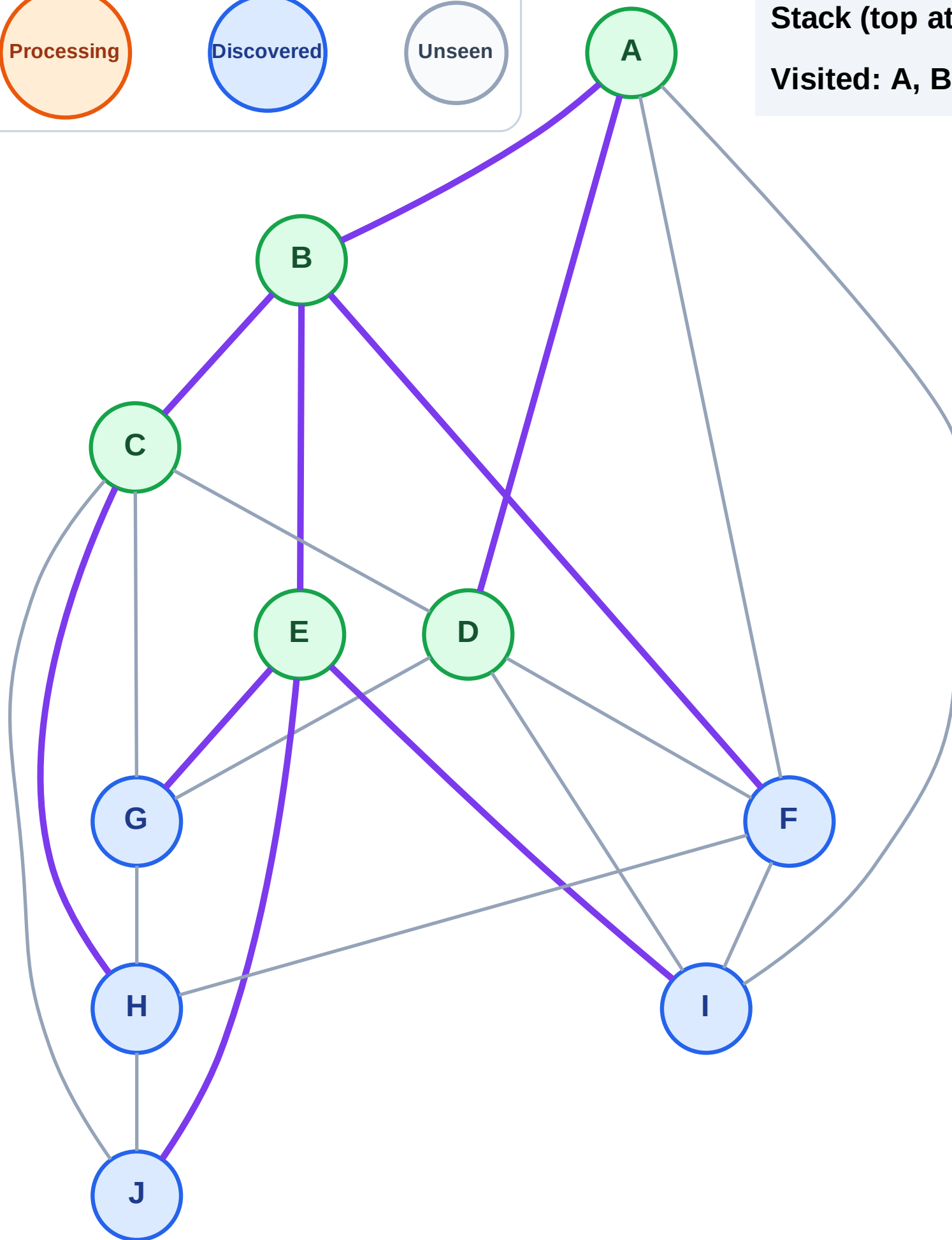
Step 11: Discover H from C

Step 11: Discover H from C



Stack (top at right): J | I | G | F | H

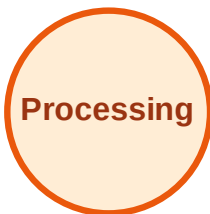
Visited: A, B, C, D, E



DFS Traversal

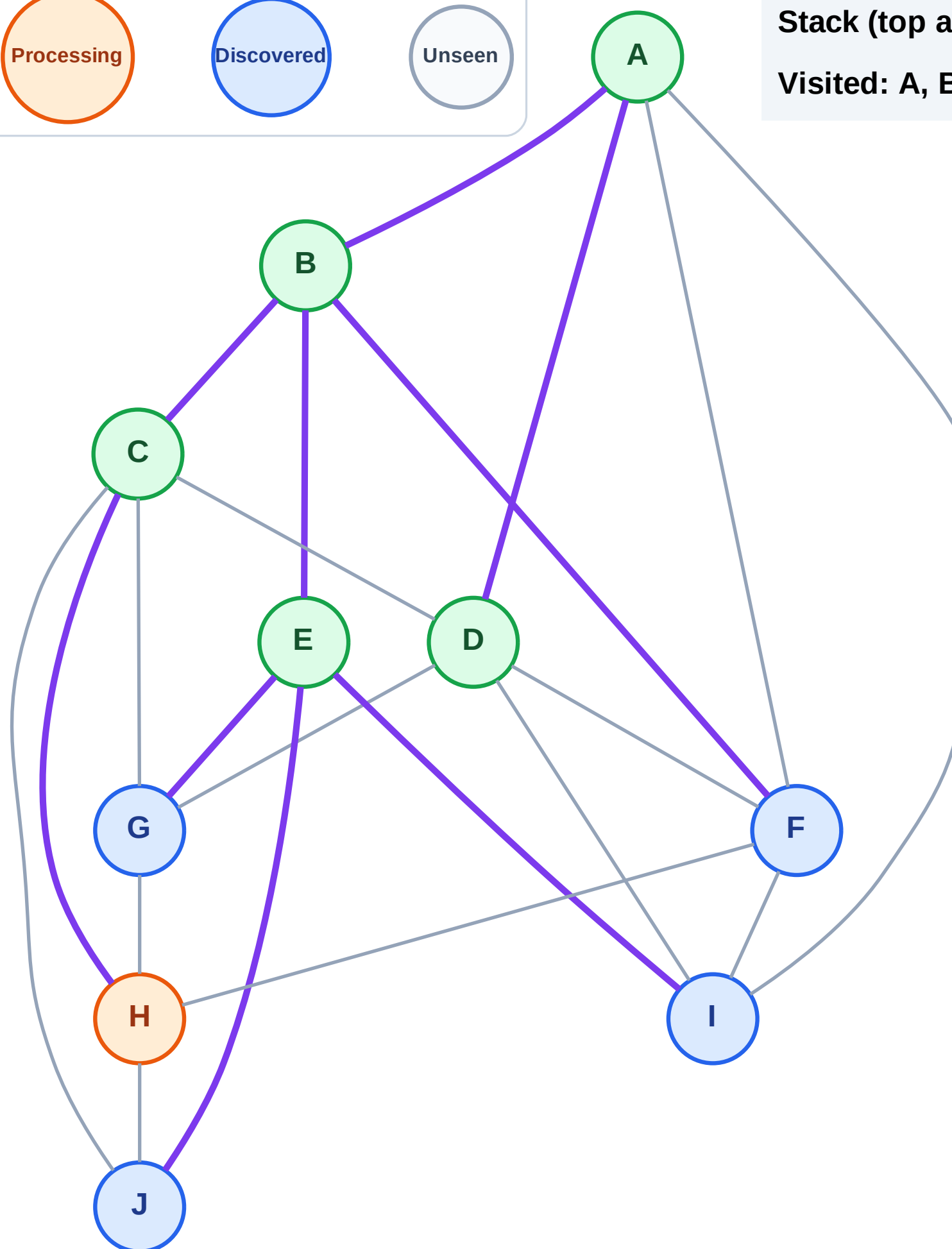
Step 12: Process node H

Legend



Stack (top at right): J | I | G | F

Visited: A, B, C, D, E



DFS Traversal

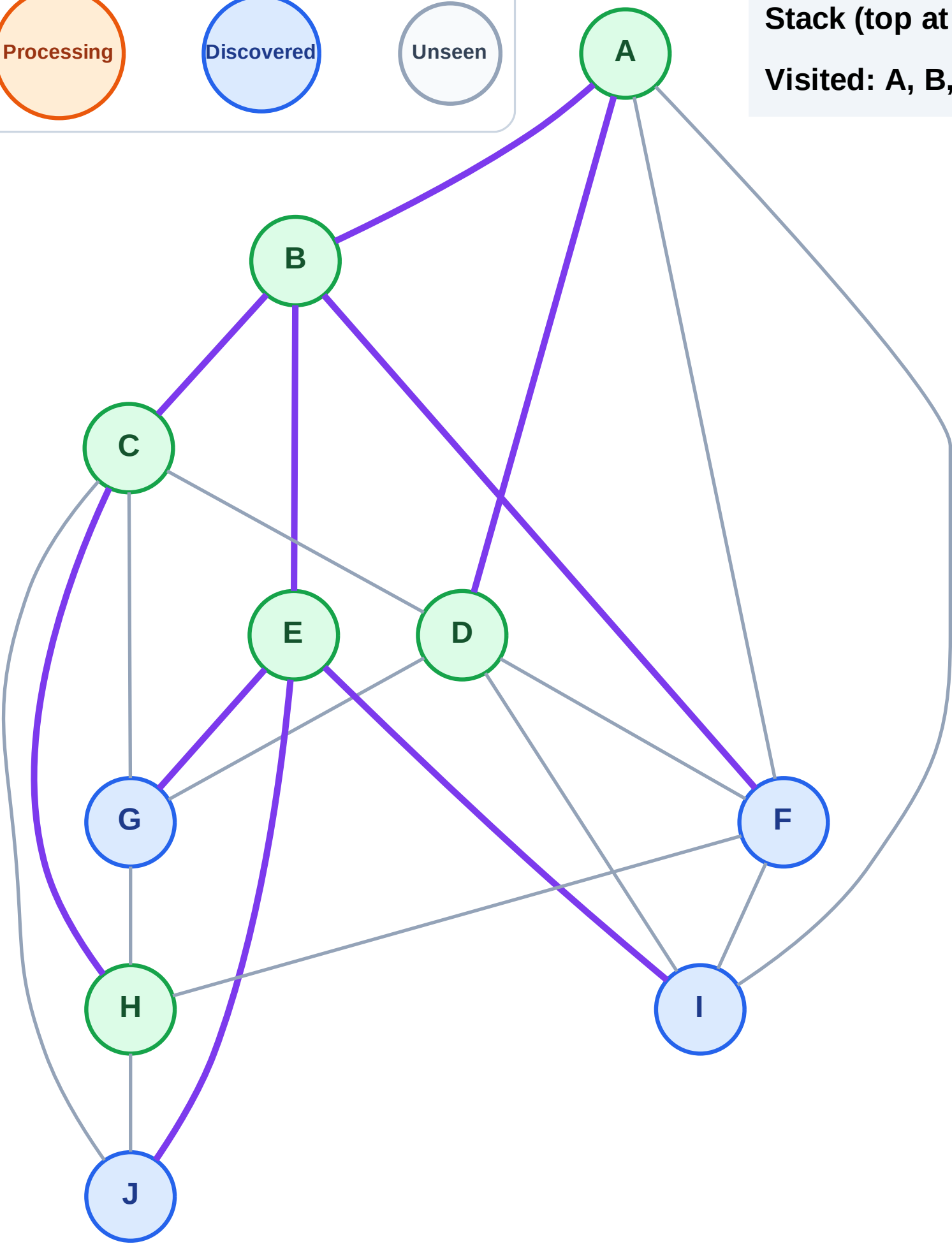
Step 13: No new neighbors from H

Legend



Stack (top at right): J | I | G | F

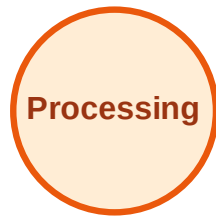
Visited: A, B, C, D, E, H



DFS Traversal

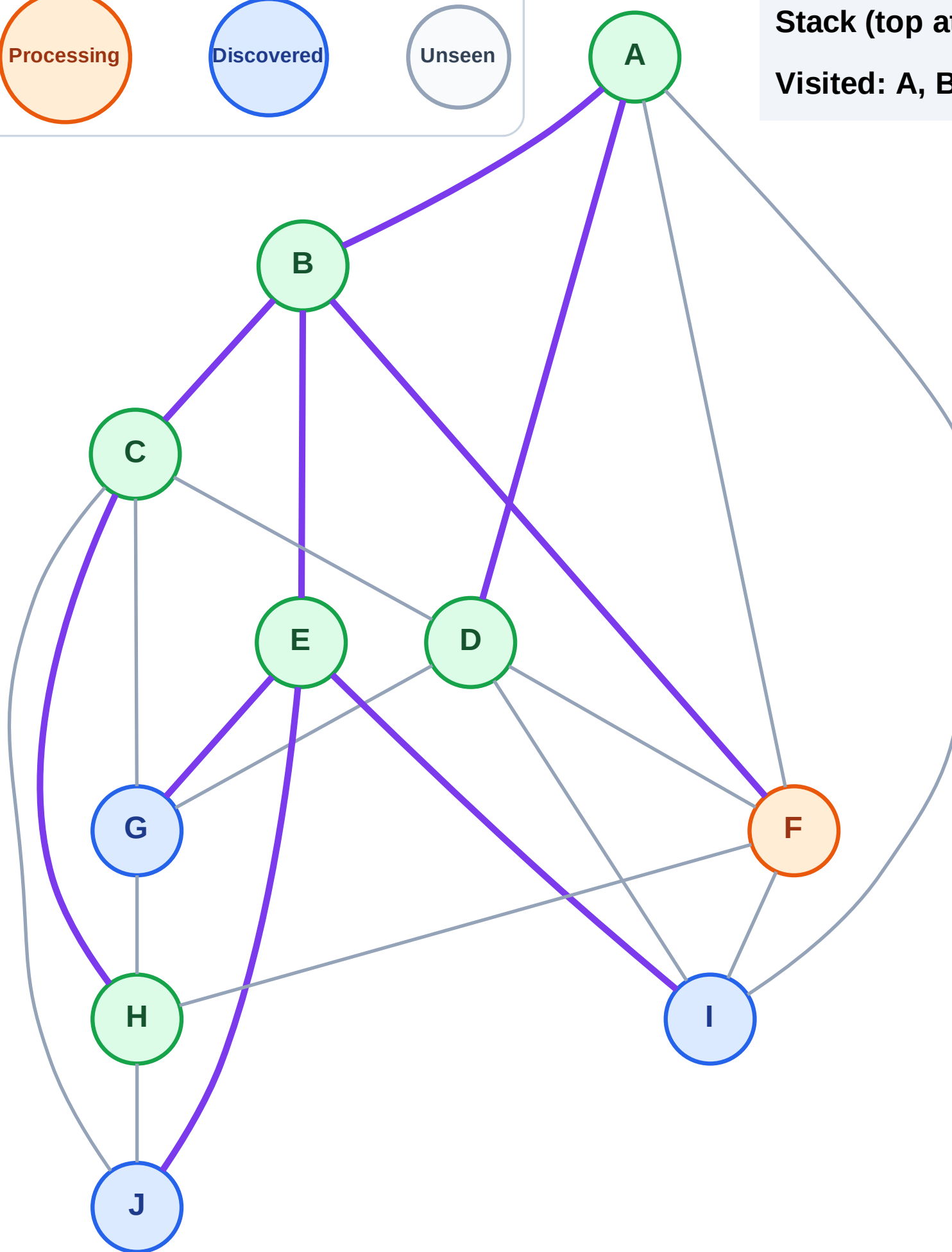
Step 14: Process node F

Legend



Stack (top at right): J | I | G

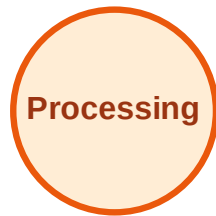
Visited: A, B, C, D, E, H



DFS Traversal

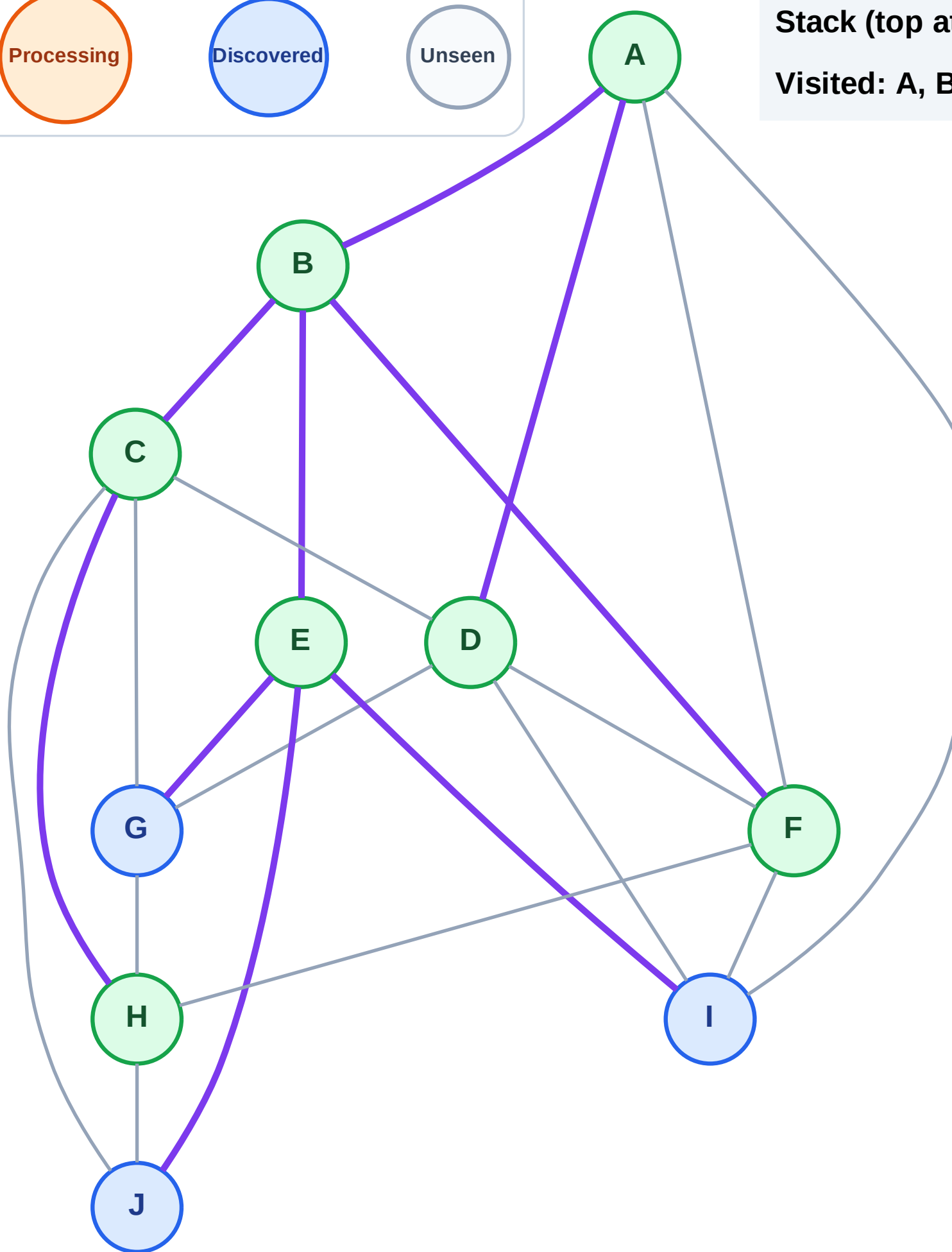
Step 15: No new neighbors from F

Legend



Stack (top at right): J | I | G

Visited: A, B, C, D, E, F, H



DFS Traversal

Step 16: Process node G

Legend

Complete

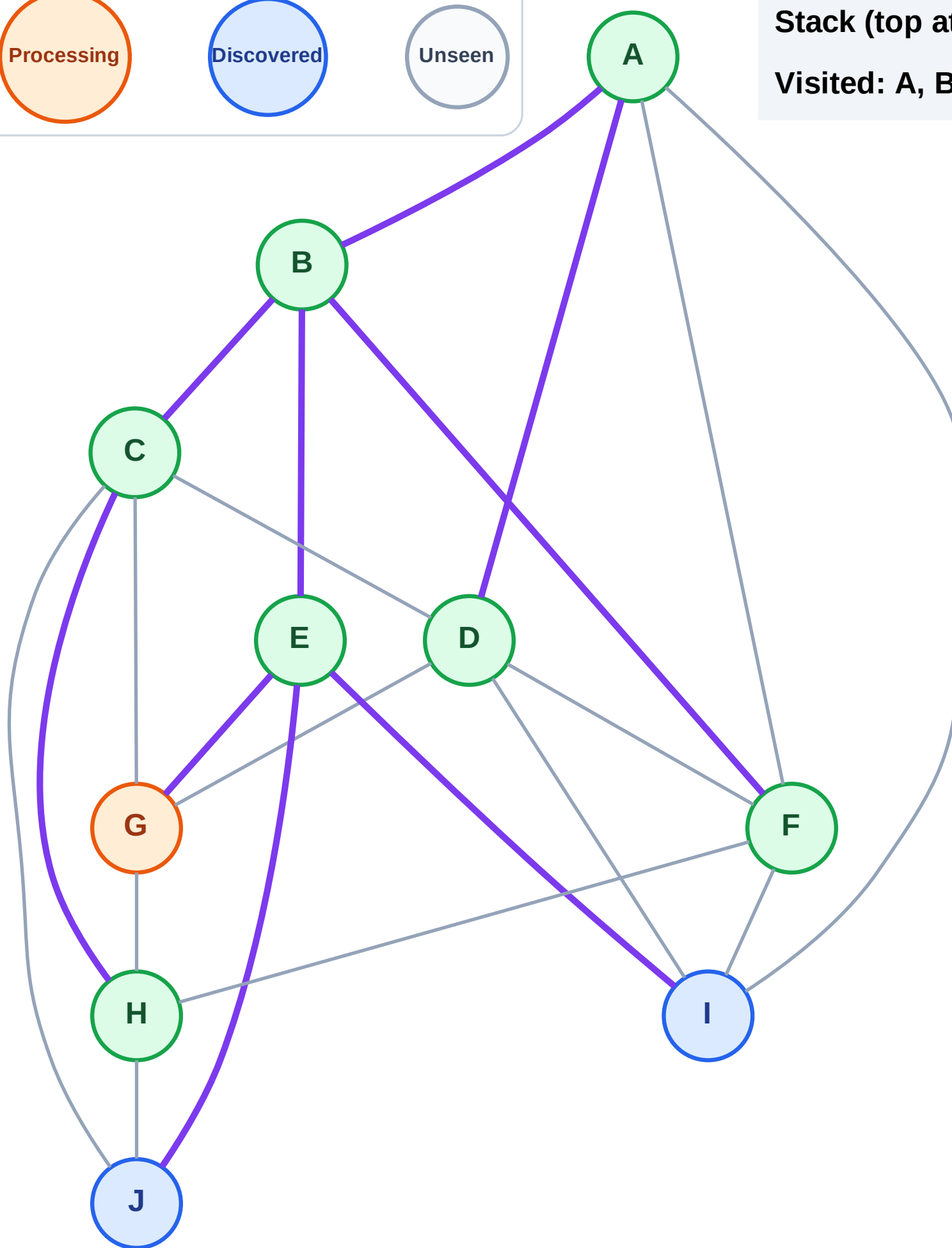
Processing

Discovered

Unseen

Stack (top at right): J | I

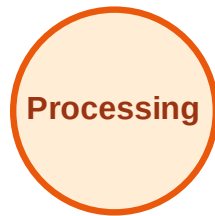
Visited: A, B, C, D, E, F, H



DFS Traversal

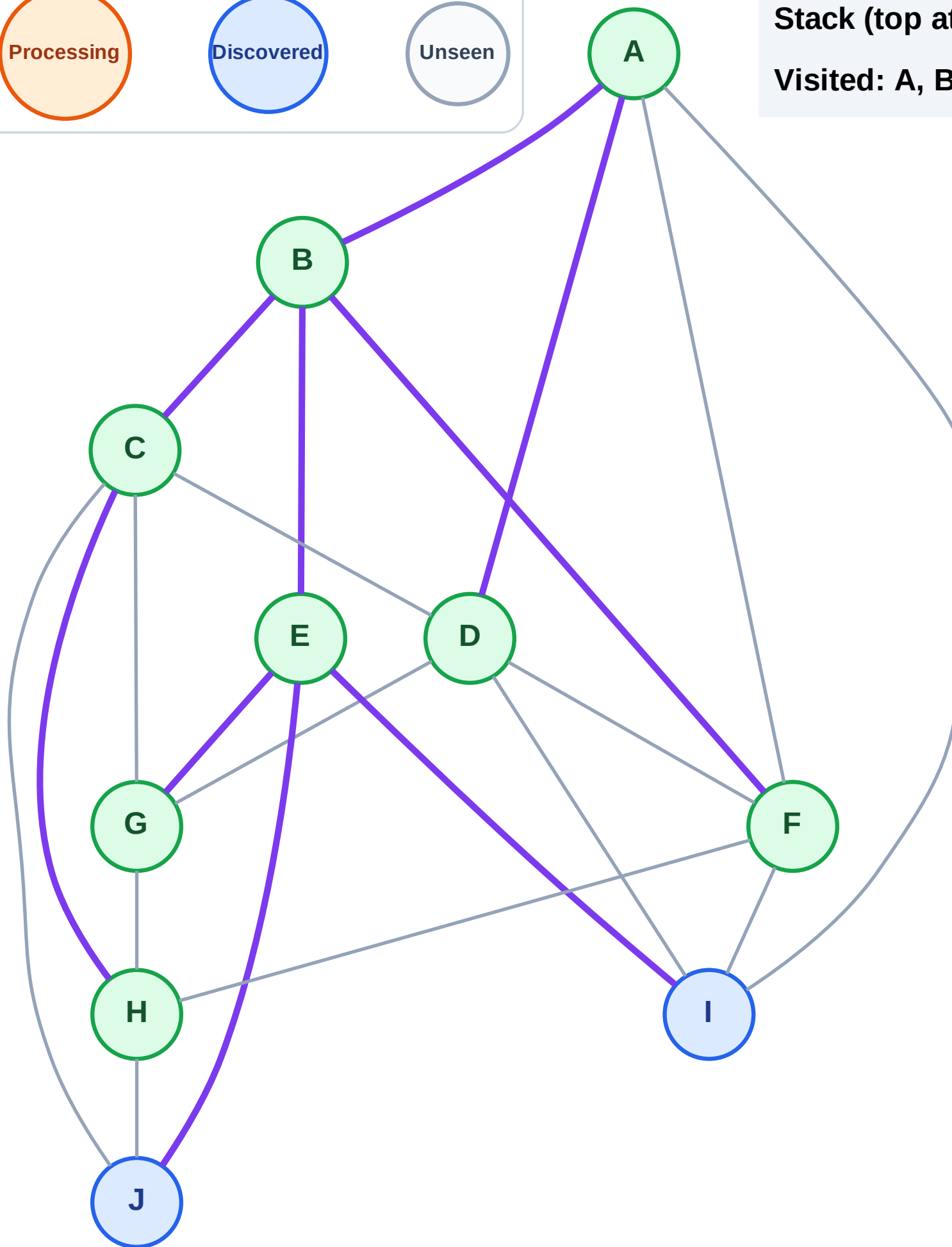
Step 17: No new neighbors from G

Legend



Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H



DFS Traversal

Step 18: Process node I

Legend

Complete

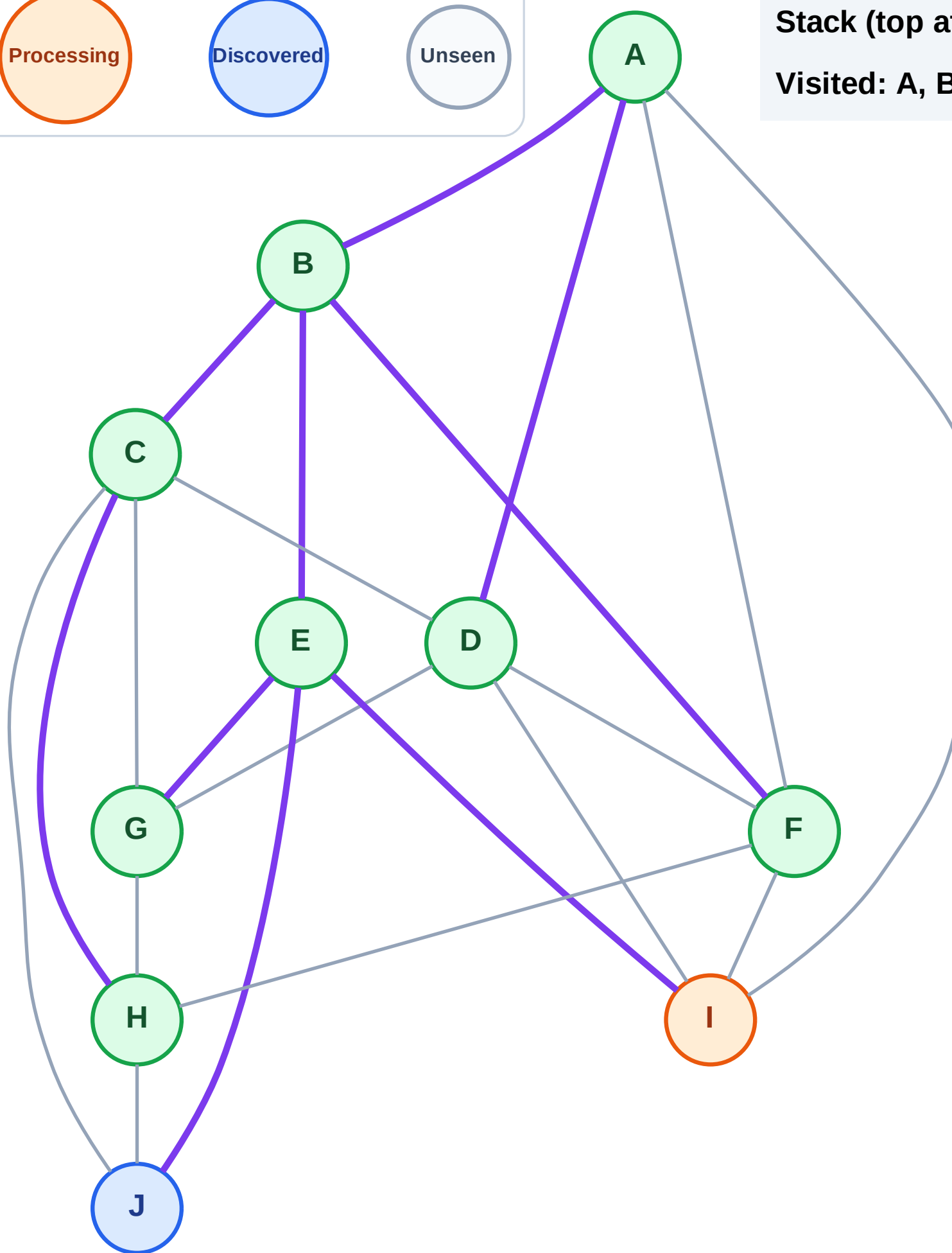
Processing

Discovered

Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H



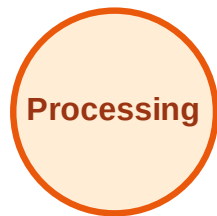
DFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



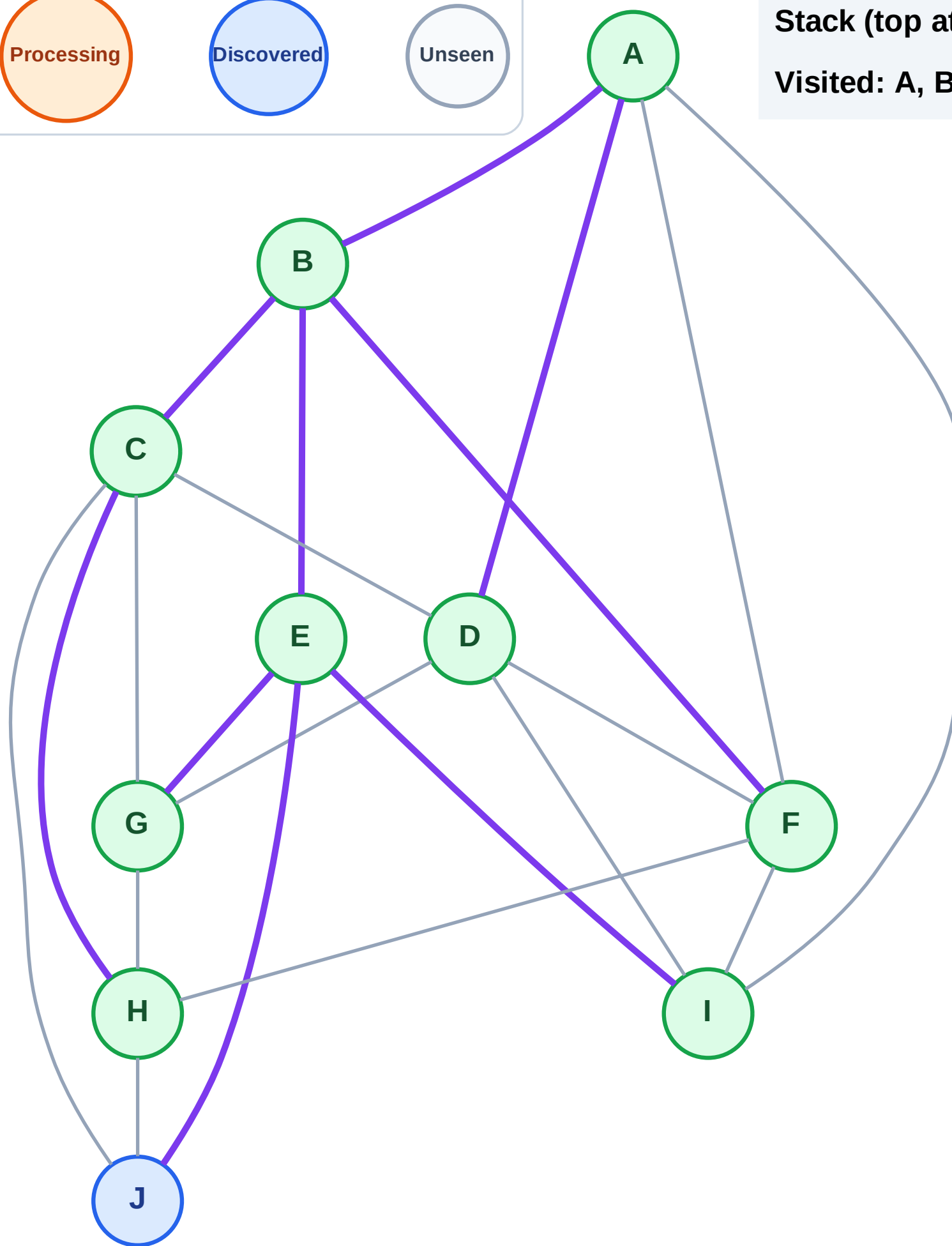
Discovered



Unseen

Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



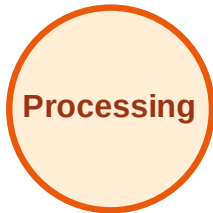
DFS Traversal

Step 20: Process node J

Legend



Complete



Processing



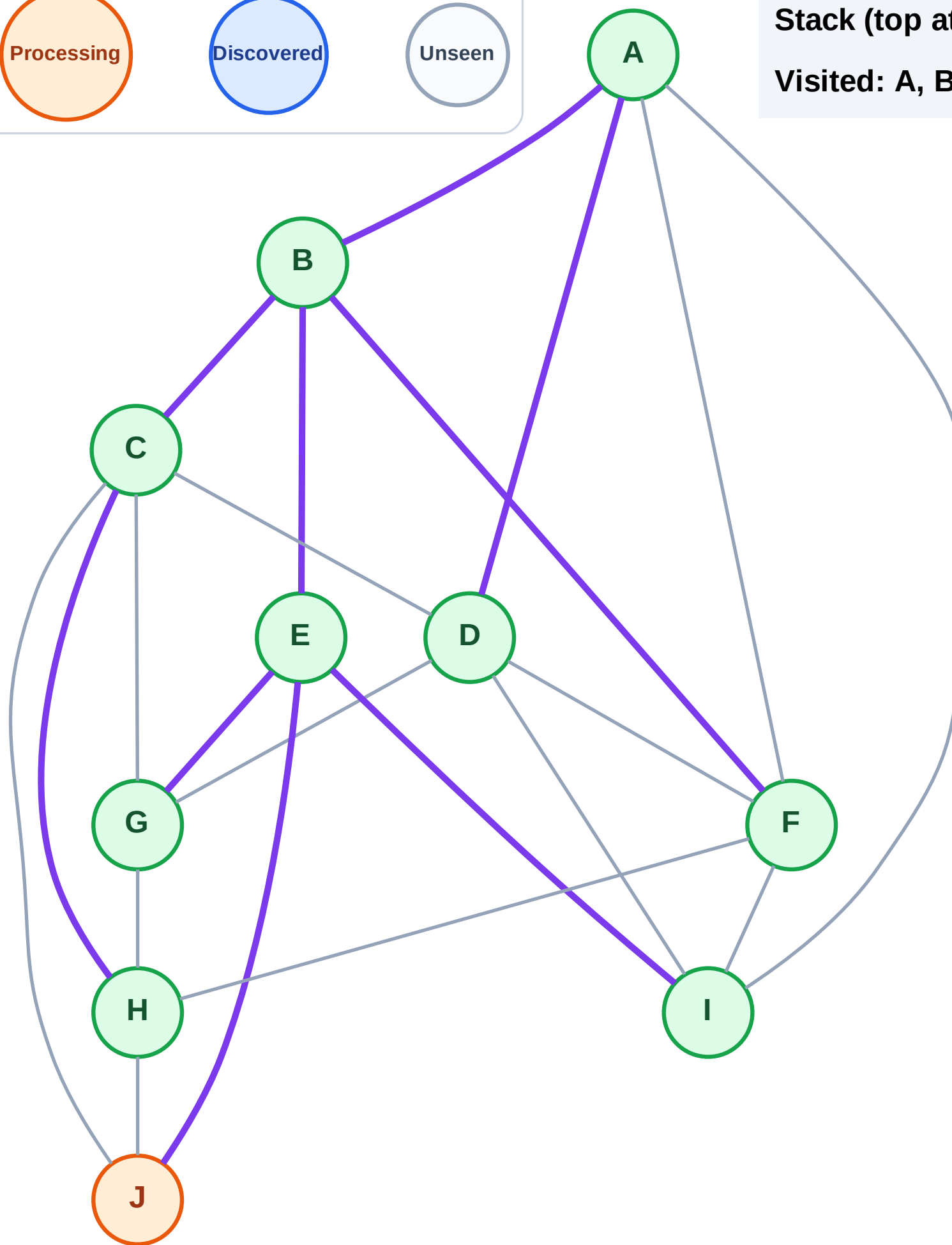
Discovered



Unseen

Stack (top at right): (empty)

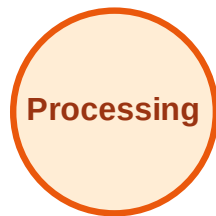
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

